# Chapter Three

# Advanced Router Configuration

# EIGRP

- The **limitations of RIP** led to the development of more advanced protocols.
- Cisco developed **EIGRP** as a proprietary distance vector routing protocol.
- It has **enhanced capabilities** that address many of the limitations of other distance vector protocols.
- EIGRP **shares some of features of RIP**, while employing many **advanced features**.

- EIGRP contains **many features** that are not found in any other routing protocols.

- All of these features **makes EIGRP an excellent choice** for **large, multi-protocol networks** that employ primarily Cisco devices.

- The two main goals of EIGRP are:-

➢ **Provide a loop-free routing environment and**

➢ **Rapid convergence.**

- To achieve these goals, EIGRP uses a **different method** for **calculating the best route**.
- The **metric** used is a composite metric that primarily considers **bandwidth** and **delay**.
- This **metric is more accurate** than **hop count** in determining the distance to a destination network.
- The **Diffusing Update Algorithm (DUAL)** used by EIGRP guarantees **loop-free operation** while it calculates routes.
- **When a change occurs** in the network topology, **DUAL synchronizes** all affected routers simultaneously.
- The administrative distance of **EIGRP is 90**.

- If a router learns routes to the **same destination** from both **RIP** and **EIGRP**, **it chooses the EIGRP** route over the route learned through RIP.

- Its maximum hop count of **255 supports** large networks.

- Unlike other distance vector protocols, **EIGRP does not send complete tables in its updates.**

- **EIGRP multicasts** partial updates about specific changes to **only those routers that need the information**, not to all routers in the area.

- To **store network information** from the updates and support rapid convergence, **EIGRP maintains multiple tables**.

- EIGRP routers keep **route and topology information** readily available in RAM so that **they can react quickly** to changes.

- EIGRP maintains three interconnected tables:
    - Neighbor table
    - Topology table
    - Routing table

# Neighbor Table

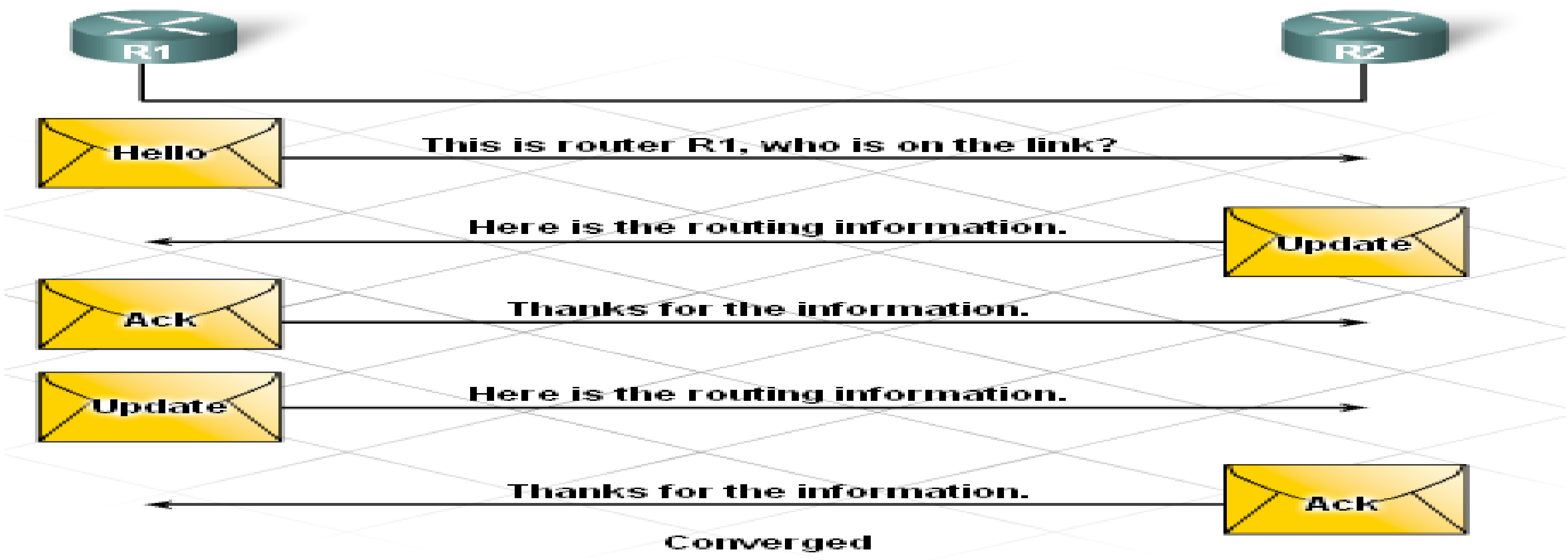
- The neighbor table lists information about **directly connected neighbor routers.**

- EIGRP **records the address** of a **newly discovered neighbor** and the **interface** that connects to it.

- When a neighbor sends **a hello packet**, it advertises **a hold time.**

- The **hold time** is the length of time that **a router treats a neighbor as reachable.**

# Cont…




After the initial exchange, routing updates are only sent when a route metric changes.

- If a **hello packet is not received** within the hold time, the timer expires and **DUAL recalculates the topology**.
- Since **fast convergence** depends on accurate **neighbor information**, this table is **crucial to EIGRP operation**

# Topology table

- The topology table **lists all routes** learned from each **EIGRP neighbor.**
- The topology table identifies up to **four primary loop-free** routes for any one destination.
- EIGRP load balances, or sends packets to a destination using **more than one path**.

- It load balances using successor routes that are **both equal cost** and **unequal cost.**
- This feature **avoids overloading** any one route with packets.

## Routing Table

- Whereas the topology table contains information about **many possible paths** to a network destination, the routing table displays only the **best paths.**

- EIGRP **displays information** about routes in two ways:

- The **routing table** labels routes learned through **EIGRP** with a **D**.

- EIGRP tags dynamic or static routes learned **from other routing** protocols as **D EX .**

# EIGRP Neighbors and Adjacencies

- Before EIGRP exchange packets between routers, it **must first discover its neighbors**.

- **EIGRP neighbors** are **other routers running EIGRP** on shared, **directly connected networks**.
- EIGRP routers use **hello packets to discover neighbors** and establish adjacencies with neighbor routers.

- The **hello packet contains** information about the **router interfaces** and the **interface addresses**.

- When a **neighbor adjacency is established**, EIGRP uses various **types of packets** to exchange and **update routing table** information.

- Neighbors learn about **new routes**, **unreachable routes**, and **rediscovered routes** through exchange of these packets:

  - Acknowledgement
  - Update
  - Query
  - Reply

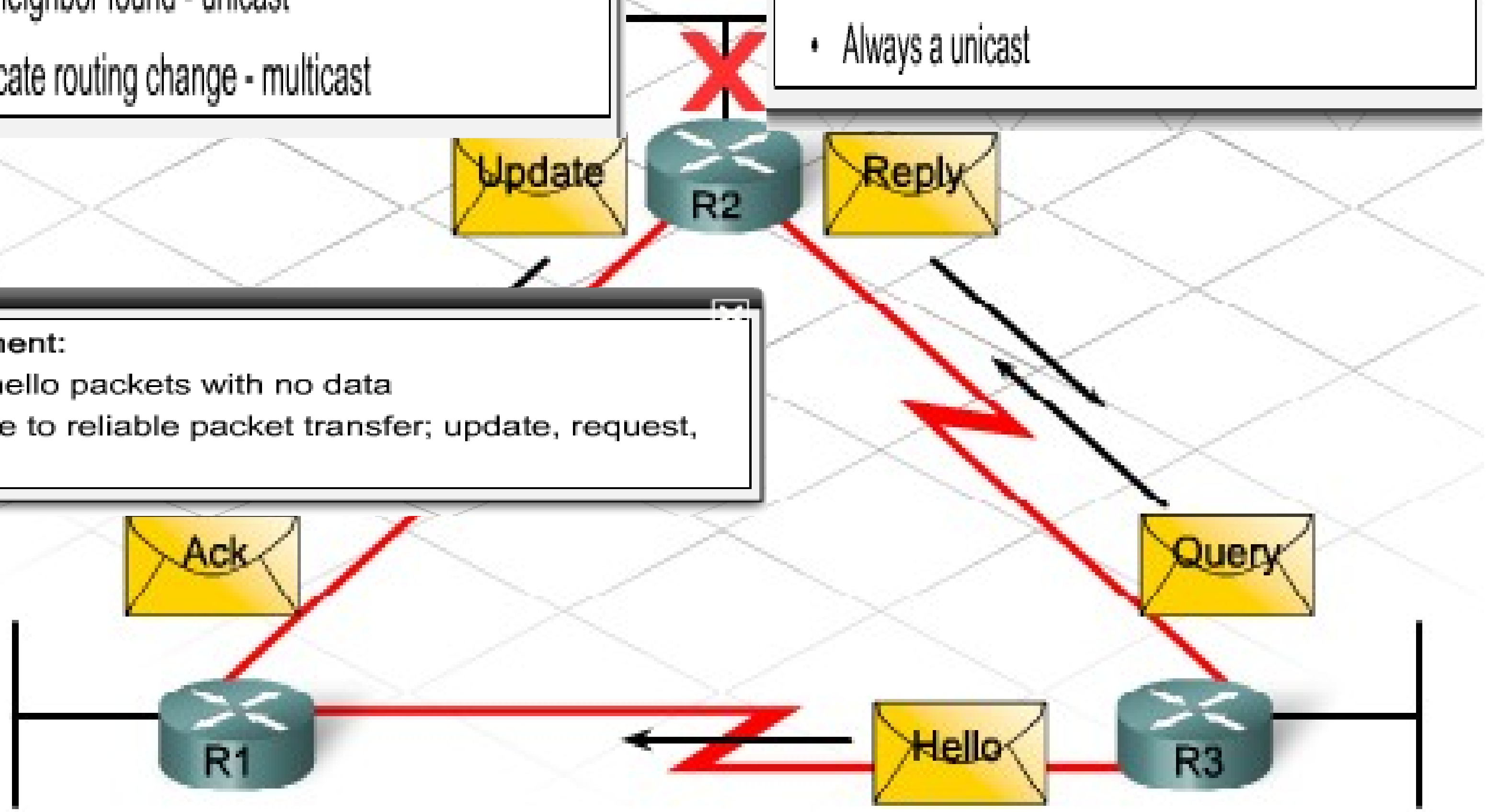
- **Acknowledgement packet:** indicates the receipt of an **update**, **query**, or **reply packet**.
  - These types of packets are **always unicast**.
- **Update packet:-** sends information about the **network topology to its neighbor.**
- **Query packet**:-message used to **inquire** about the value of some variable.
  - Whenever DUAL places a route in the active state, the router must send a **query packet** to each neighbor.
- **Replay packet**:-Information sent when a query packet is received.
  - Neighbors must send replies, even if the reply states that no information on the destination is available.
  - Queries can be multicast or unicast. Replies are always unicast.

Update:

- If new neighbor found - unicast
- To indicate routing change - multicast

Reply:

- Response to a query
- Always a unicast




Acknowledgement:

- Unicast hello packets with no data
- Response to reliable packet transfer; update, request, reply

Query:

- To request specific info about a neighbor or multicast looking for new successor
- Can be multicast or unicast

Roll over each p

# EIGRP configuration

- To begin the EIGRP routing process, use two steps:

**Step 1**

➢ Enable the EIGRP routing process.

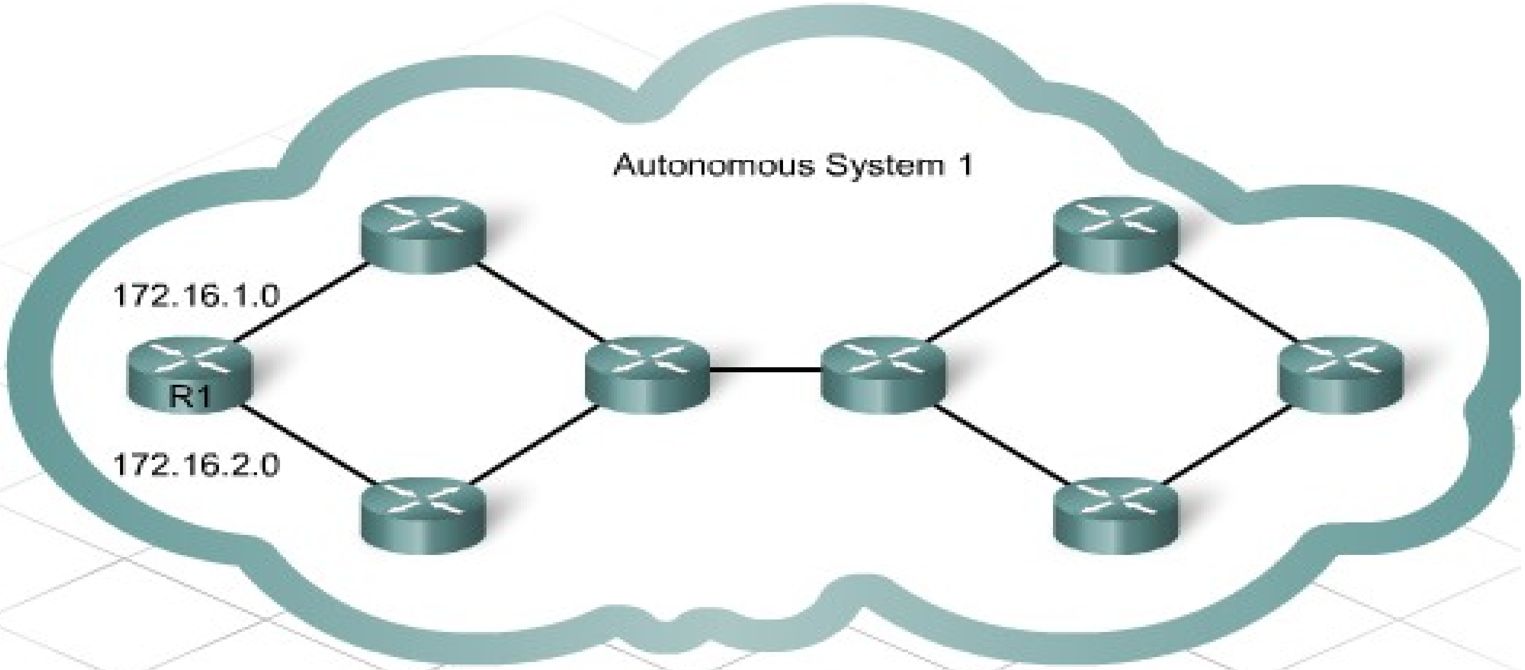
- Although EIGRP refers to the parameter as an autonomous system number, it actually functions as a process ID.

Step 1

```
R1(config)#router eigrp ?
   <1-65535>   Autonomous system number
R1(config)#router eigrp 1
```

# Step 2

- Include network statements for each network to be advertised.
- The network command tells EIGRP **which networks** and **interfaces participate** in the EIGRP process.




Step 1

```
R1(config)#router eigrp ?
   <1-65535>   Autonomous system number
R1(config)#router eigrp 1
```

Step 2

```
R1(config-router)#network 172.16.0.0
```

# Issue and Limitation of EIGRP

- Does not work in **a multi-vendor** environment because it is a **Cisco proprietary protocol**
- Must share the **same autonomous** system among routers and cannot be **subdivided into groups.**
- Create very **large routing tables**, which requires large update packets and **large amounts of bandwidth.**
- Uses **more memory** and processor power than RIP
- Requires administrators with **advanced technical knowledge** of the protocol and the network

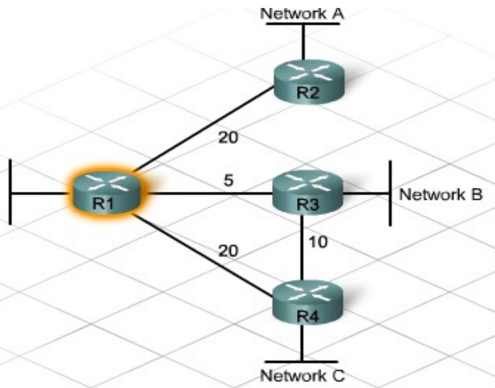
# Open Shortest Path First (OSPF)

- Open Shortest Path First (OSPF) is an example of a **link-state routing protocol.**
- It divides the network into different sections, which are referred to **as areas**.
- This division allows for greater **scalability.**
- OSPF, **do not send frequent** periodic updates of the entire routing table.

- **Compared with distance vector protocols**, link-state routing protocols:

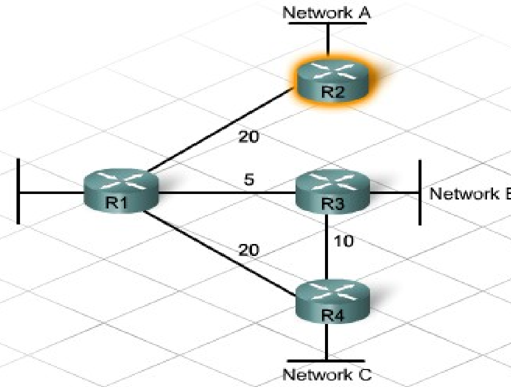
- Requires more network **planning and configuration**
- Requires increased **router resources**
- Requires more **memory** for storing multiple tables
- Requires **more CPU** and processing power for the complex routing calculations

- OSPF routers within **a single area advertise** information about the status of their links **to their neighbors**.
- Messages called **Link State Advertisements (LSAs)** are used to advertise this status information.
- It uses the algorithm called **Dijkstra's Algorithm**, to generate a topological tree, or map of the network.
  - Convergence occurs when all routers
    - ➢ Receive information about **every destination** on the network
    - ➢ Process this information with the **SPF algorithm**
    - ➢ Update their **routing tables**



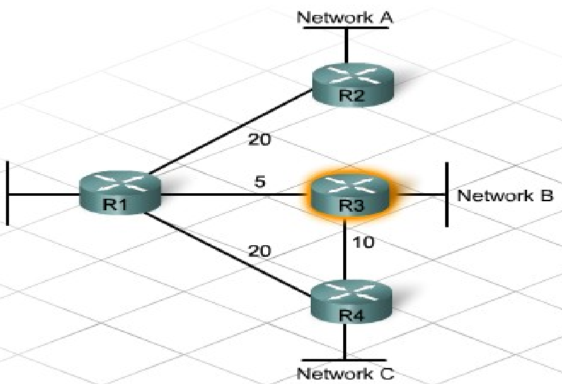

| Destination | Routes | Cost |
|---|---|---|
| A | R1 - R2 | 20 - Least Cost Path |
| B | R1 - R3 | 5 - Least Cost Path |
| | R1 - R4 - R3 | 30 |
| C | R1 - R3 - R4 | 15 - Least Cost Path |
| | R1 - R4 | 20 |

Click each router to view the SPF tree and paths.

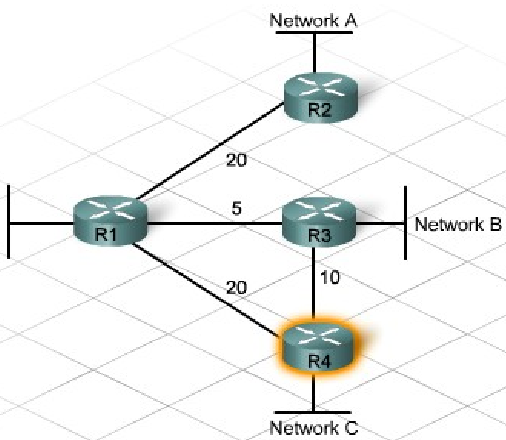



| Destination | Shortest Path | Cost |
|---|---|---|
| A | Connected | 0 - Least Cost Path |
| B | R2 - R1 - R3 | 25 - Least Cost Path |
| | R2 - R1 - R4 - R3 | 50 |
| C | R2 - R1 - R3 - R4 | 35 - Least Cost Path |
| | R2 - R1 - R4 | 40 |

Click each router to view the SPF tree and paths.




| Destination | Shortest Path | Cost |
|---|---|---|
| A | R3 - R1 - R2 | 25 - Least Cost Path |
| | R3 - R4 - R1 - R2 | 50 |
| B | Connected | 0 - Least Cost Path |
| C | R3 - R4 | 10 - Least Cost Path |
| | R3 - R2 - R4 | 25 |




| Destination | Shortest Path | Cost |
|---|---|---|
| A | R4 - R3 - R1 - R2 | 35 - Least Cost Path |
| | R4 - R1 - R2 | 40 |
| B | R4 - R3 | 10 - Least Cost Path |
| | R4 - R1 - R3 | 25 |
| C | Connected | 0 - Least Cost Path |

- **DR router**:-It receive LSA contain routing information from neighbor router, and it send to **all other routers**.
- **BDR router**:-router that is identified to take over **the DR router fails**

- The **purpose of the DR and BDR** is to **reduces** the number of **updates sent**, unnecessary traffic flow, and **processing overhead on all routers.**

- The router with the highest **router ID** is elected the **DR**.

- The **second highest** is elected as the B**DR**.

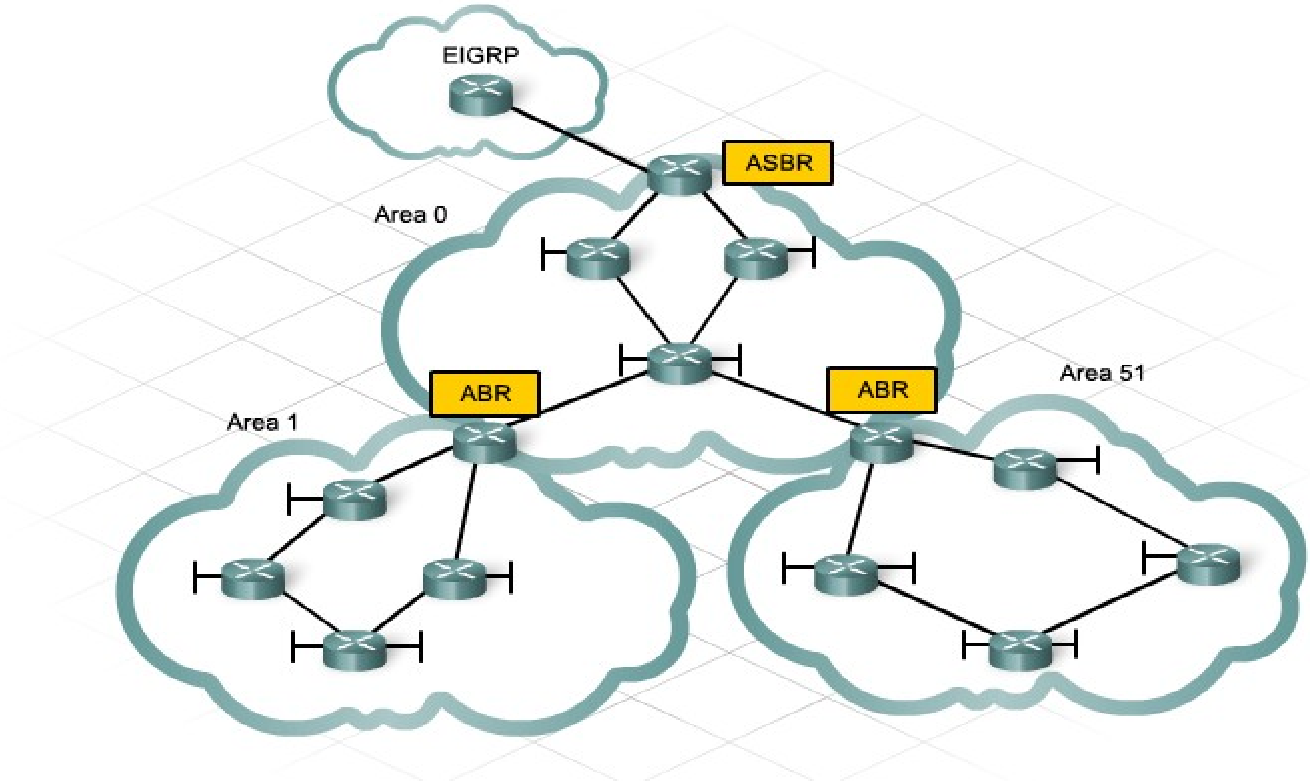
- By default, OSPF routers have a priority **value of 1**.
- The highest value that can be set for router priority is **255**.
- A value of **0** signifies that the router is **ineligible** to be DR or BDR.
- An **administrator** can configuring **a priority using** the interface configuration command:

**ip ospf priority number**

# OSPF Area

- All OSPF networks begin with Area 0, also called the **backbone area.**
- As the network is expanded, **other areas** can be created that are **adjacent to Area 0**.
- These other areas can be assigned any number, up to **65,535**.
- The maximum number of routers allowed in **one area is 50**.
- OSPF has a **two-layer hierarchical design**.
- **Area 0**, also referred to as the **backbone area**, exists at the **top** and all other areas are located at the **next level.**
- All **non-backbone areas** must **directly connect** to area 0.
- A **group of areas** creates an OSPF **Autonomous System (AS)**.

- A router that **connects an area** to the **backbone area** is called an **Area Border Router (ABR).**
- A router that connects an area to a different routing protocol, such as EIGRP called an **Autonomous System Border Router (ASBR).**



# OSPF Configuration

- Configuration of basic OSPF is not a complex task, it requires only two steps.
- The first step enables the OSPF routing process.
- The second step identifies the networks to advertise.
- **Step 1: Enable OSPF**
- router(config)#router ospf <process-id>
- The process ID is chosen by the administrator and can be any number from 1 to 65535.
- **Step 2: Advertise networks**
- Router(config-router)#network <network-address> <wildcard-mask> area <area-id>

- The OSPF network statement requires, the **wildcard** mask is the **inverse of the subnet mask**.
- To determine the wildcard mask for a network subtract the decimal subnet mask the all 255s mask (255.255.255.255).

All 255s mask: 255.255.255.255

Subnet mask: -255.255.255.0

**Wildcard mask**: 0 . 0 . 0 .255

**Router(config-router)#network 192.168.10.0 0.0.0.255 area 0**

# BGP(Border Gateway Protocol )

- **Exterior routers** exchange information about how to reach various networks using **exterior protocols**.
- Exterior routing protocols **seek to find the best path** through the **Internet** as a sequence of Autonomous Systems.
- The most common exterior routing protocol on the Internet today is **Border Gateway Protocol (BGP)**

# Cont…

- Packets are routed across the Internet in several steps:

1. The source host sends a packet to host located in another AS.

2. Since the destination IP address of the packet is not a local network, the interior routers keep passing the packet, until it arrives at an **exterior router at the edge of the local AS**.

3.The exterior router **maintains a database** for all the Autonomous Systems with which it connects.

4. The **exterior router directs** the packet to its next hop on the path, which is the exterior router at the **neighboring AS**.

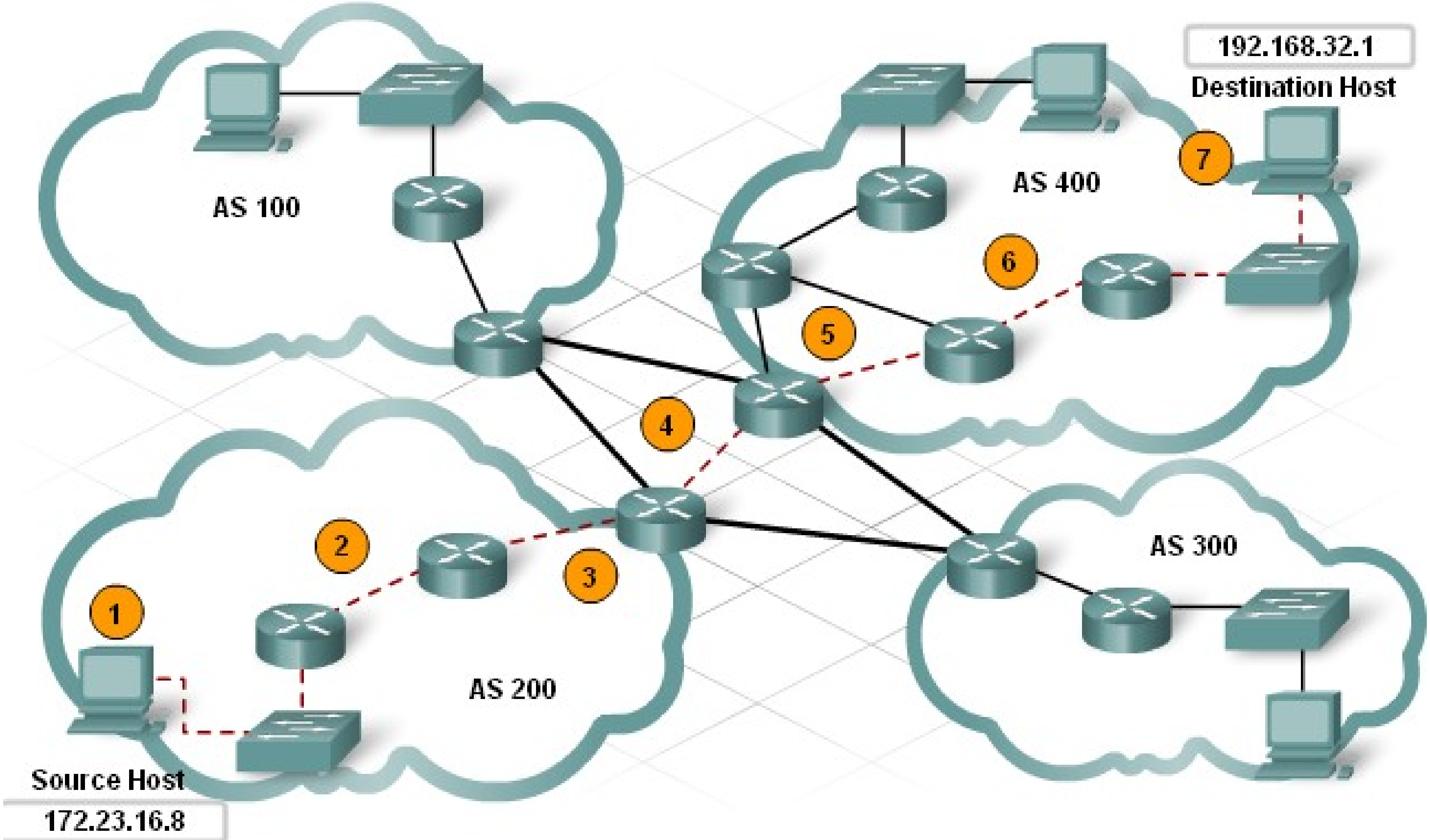
# Cont…

5. The **packet arrives at the neighboring AS**, where the exterior router checks its own reachability database and forwards the packet to **the next AS on the path**.

6. The **process is repeated at each AS until the exterior router at the destination AS** recognizes the destination IP address of the packet as an internal network in that AS.

7.The **final exterior router then directs the packet to the next hop** interior router listed in its routing table. From then on, the packet is treated just like any local packet.

# Cont…



192.168.32.1
Destination Host
AS 100
AS 400
AS 200
AS 300
1
2
3
4
5
6
7
Source Host
172.23.16.8

# Cont…

- The first step in **enabling BGP** on a router is to configure the AS number.

  **router bgp [AS number]**

- The next step is to **identify the ISP router that is the BGP** neighbor with which the Customer Premise Equipment (CPE) router exchanges information. The command to identify the neighbor router is:

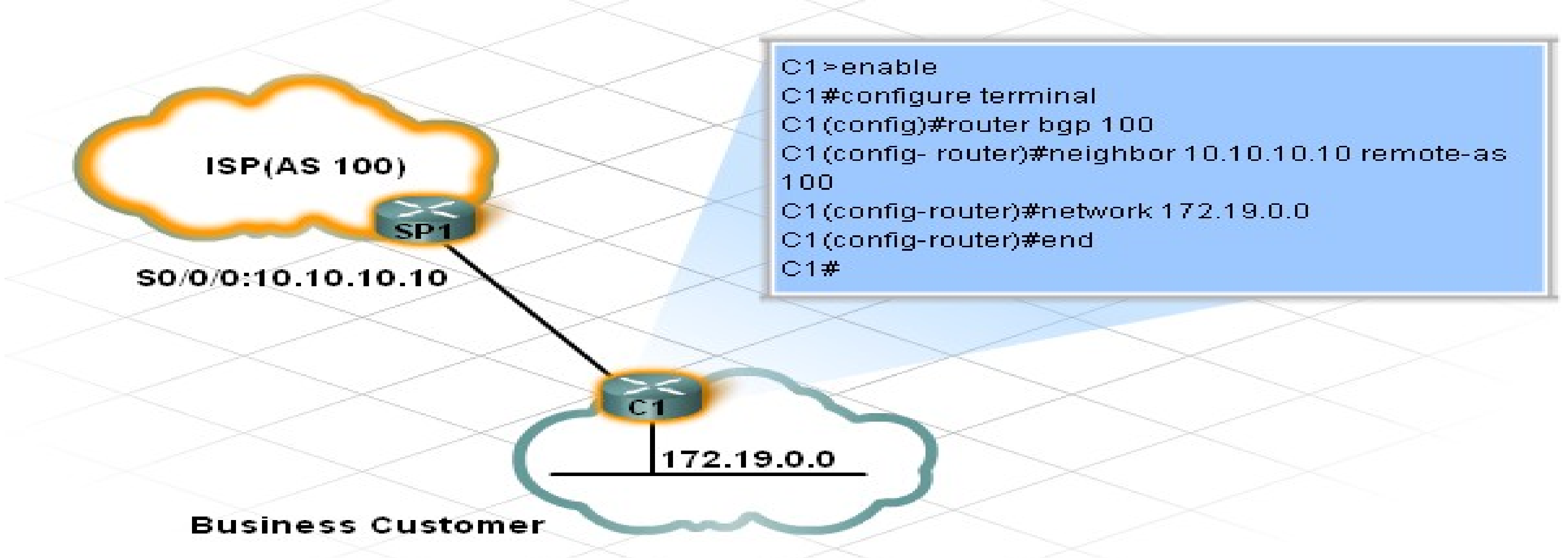
  **neighbor [IP Address] remote-as [AS number]**

- To use BGP to advertise an internal route, a network command is needed. The format of the network command is:

  **network [network address]**

# Cont….

- Once all of the customer premise equipment is installed and the **routing protocols configured**, the customer has both local and Internet connectivity.
- Now the **customer is able to fully participate** in other services the ISP may offer.
- The IP addresses used for BGP are normally registered, routable addresses which identify unique organizations.



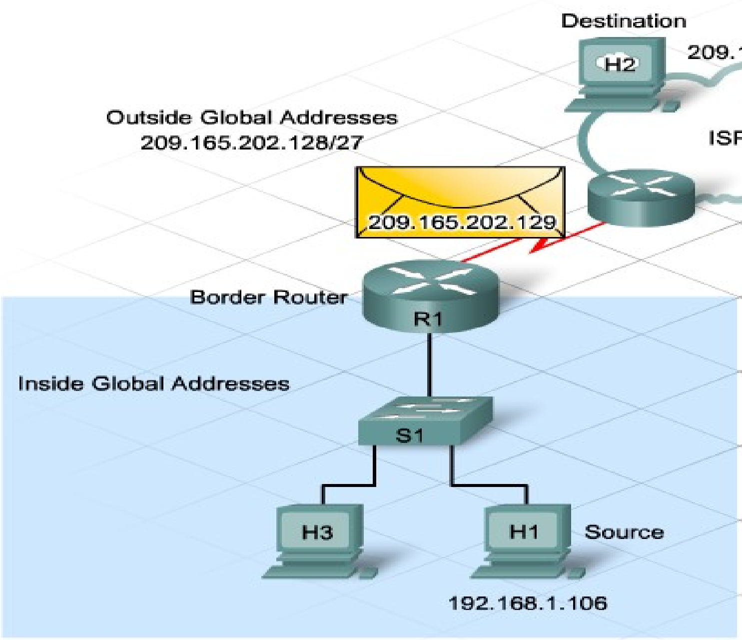
# NAT (Network address translation )

- Private addresses are available for anyone to use in their enterprise networks.
- Class A: 10.0.0.0 - 10.255.255.255
- Class B: 172.16.0.0 - 172.31.255.255
- Class C: 192.168.0.0 - 192.168.255.255

Using **private addressing has these benefits**:

- It save company not **purchased many public addresses** for each host.
- It allows **thousands of internal employees** to use a **few public addresses.**
- It **provides a level of security**, because users from other networks or organizations cannot see the internal addresses.

- **NAT translates** internal private addresses into one or more **public addresses** for routing onto the Internet.
- Organizations **connect to their ISPs** through a **single con**nection.
- The **local boundary** router configured with **NAT** connects to the ISP.
- NAT is enabled on **one device** (normally the border or edge router)

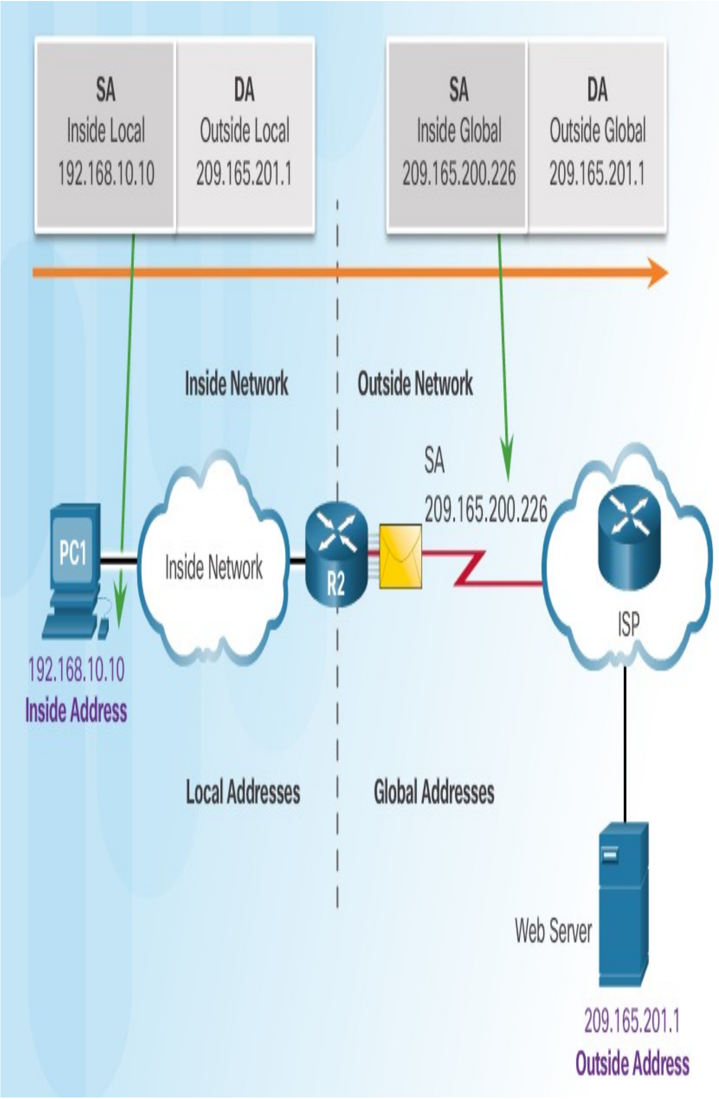



| | |
|---|---|
| H1 before translation | 192.168.1.106 |
| H1 after translation | 209.165.202.129 |

# NAT Terminology

- Four types of addresses: **inside, outside, local,** and **global**

- **Inside address** – address of the company network device that is being translated by NAT
- **Outside address** – IP address of the destination device
- **Local address** – any address that appears on the inside portion of the network
- **Global address** – any address that appears on the outside portion of the network

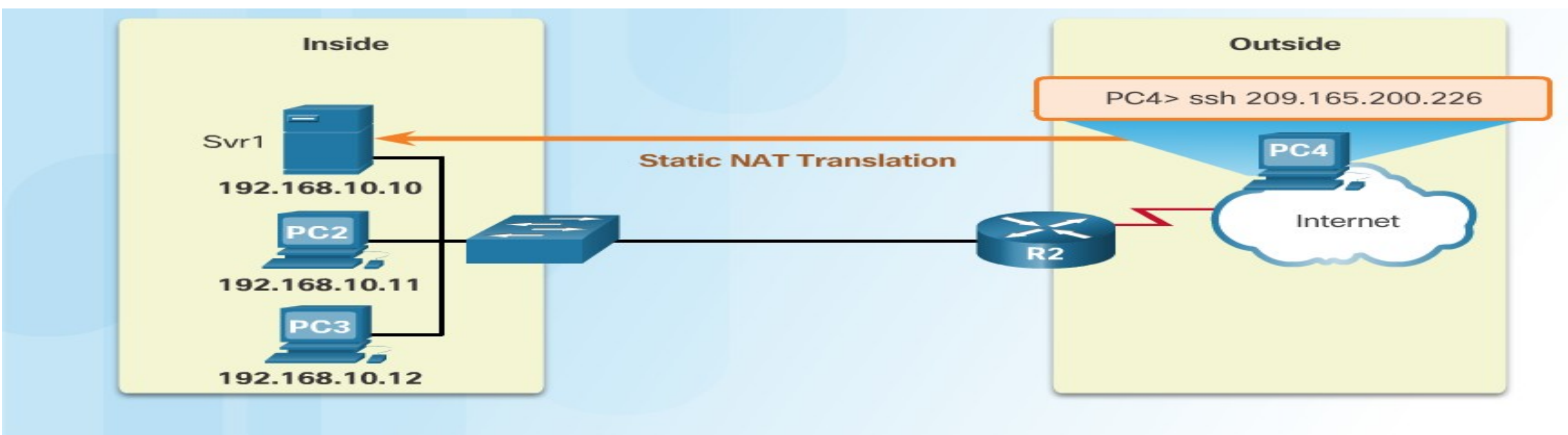


# Static NAT

- NAT can be configured statically or dynamically.

❑ **Static NAT**:- An internal host mapped with fixed public IP all of the time.

❑ **One-to-one address** mapping between local and global addresses




**Static NAT Table**

| Inside Local Address | Inside Global Address – Addresses reachable via R2 |
|---|---|
| 192.168.10.10 | 209.165.200.226 |
| 192.168.10.11 | 209.165.200.227 |
| 192.168.10.12 | 209.165.200.228 |

## ❑Configuring Static NAT

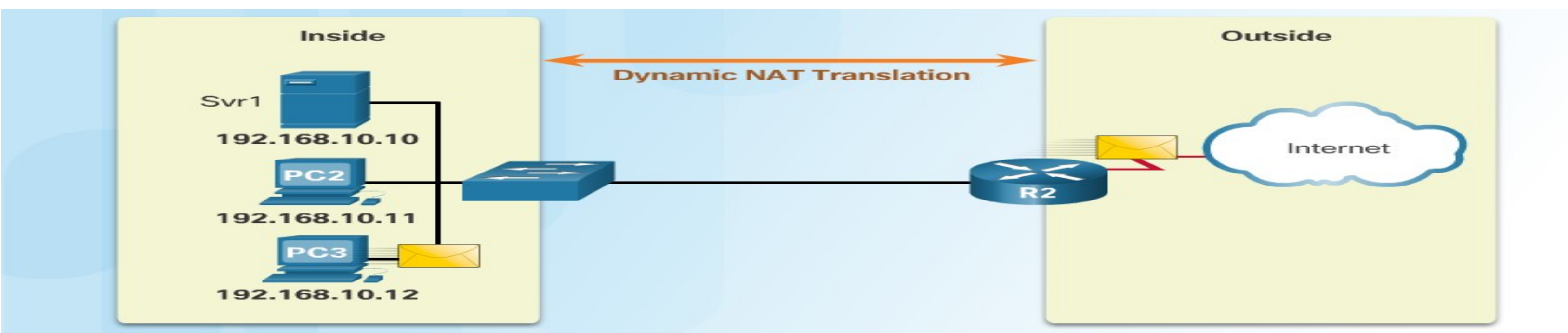
- 1. Determine the public IP address and map the inside, or private address to the public address.
- 2. Configure the inside and outside interfaces.

- Router(config)#ip nat inside source static [inside local ip address] [inside global IP address]
- Router(config)#ip nat inside source static 10.0.0.10 50.0.0.10
- Router(config-if)#ip nat inside
- Router(config-if)#ip nat outside

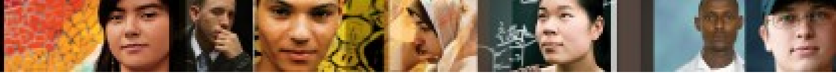
# Dynamic NAT

- **Dynamic NAT:-** uses an **available pool** of Internet public addresses and assigns them to **inside local addresses**.
- That **host uses** the assigned **public IP address** throughout the length of the session. **Once the session** ends, the **public IP address returns** to the pool for **use by another host**.
- Addresses are assigned on a first-come, first serve basis




IPv4 NAT Pool

| Inside Local Address | Inside Global Address Pool – Addresses reachable via R2 |
|---|---|
| 192.168.10.12 | 209.165.200.226 |
| Available | 209.165.200.227 |
| Available | 209.165.200.228 |
| Available | 209.165.200.229 |
| Available | 209.165.200.230 |



- When configuring either static or dynamic NAT.

➢ List any **servers** that require a permanent **outside addr**ess.

➢ Determine which internal **hosts require translation**.

➢ Determine which **interfaces source** the **internal traffic**. These will become the inside interfaces.

➢ Determine which **interface sends traffic to the Internet**. This will become the outside interface.

➢ Determine the **range of public addresses available**.

## Configuring Dynamic NAT

- 1. Identify the pool of public IP addresses available for use.
- 2. Create an access control list (ACL) to identify hosts that require translation.
- 3. Assign interfaces as either inside or outside.
- 4. Link the access list with the address pool.

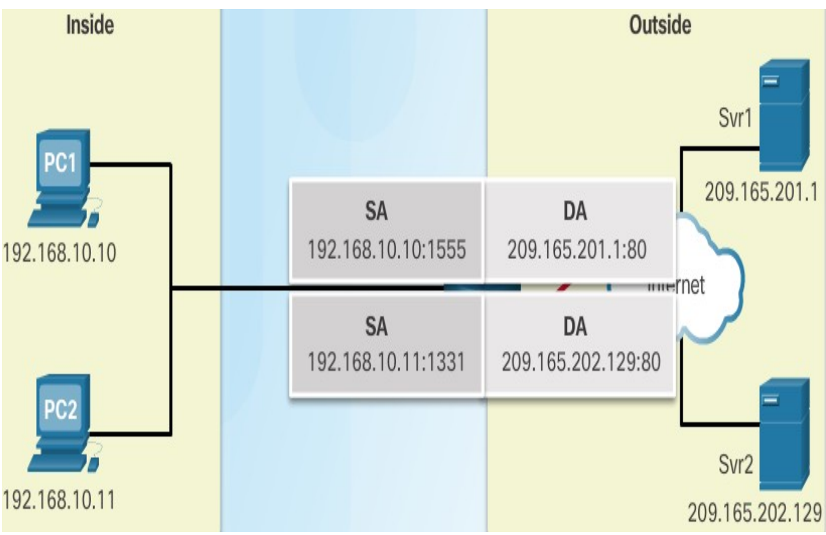
- R1(config)#access-list 1 permit 10.0.0.10 0.0.0.0
- R1(config)#access-list 1 permit 10.0.0.20 0.0.0.0
- R1(config)#ip nat pool ccna 50.0.0.1 50.0.0.2 netmask 255.0.0.0
- R1(config)#ip nat inside source list 1 pool ccna
- R1(config)#interface FastEthernet 0/0
- R1(config-if)#ip nat inside
- R1(config-if)#exit
- R1(config)#interface Serial0/0/0
- R1(config-if)#ip nat outside
- R1(config-if)#exit

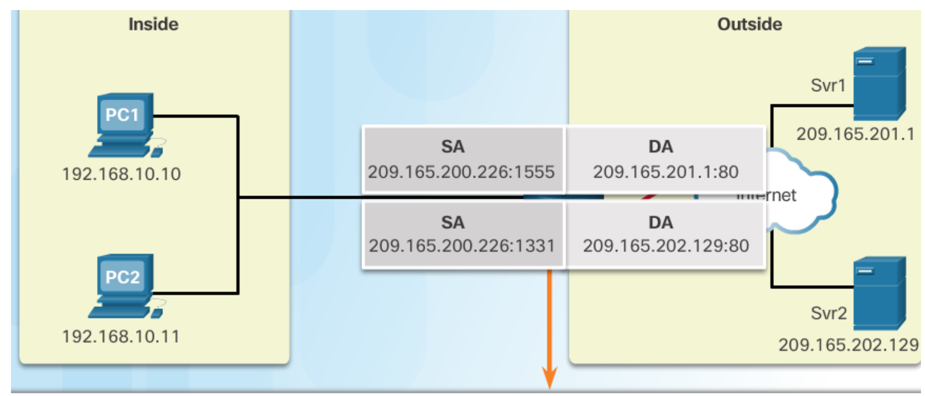
# PAT port address translation

- One of the more popular variations of **dynamic NAT** is known as **Port Address Translation (PAT),** also referred to as **NAT Overload.**
- PAT dynamically translates **multiple inside local** addresses to a **single public address**.
- In PAT router translates the **local source address** and port number combination to a **single global IP address**.
- The port number associated with the conversation is **unique**.
- We have more than 65 thousand port

# PAT

- PAT (otherwise known as NAT overload) can use one public IPv4 address to allow thousand of private IPv4 addresses to communicate with outside network devices.
- Uses port numbers to track the session



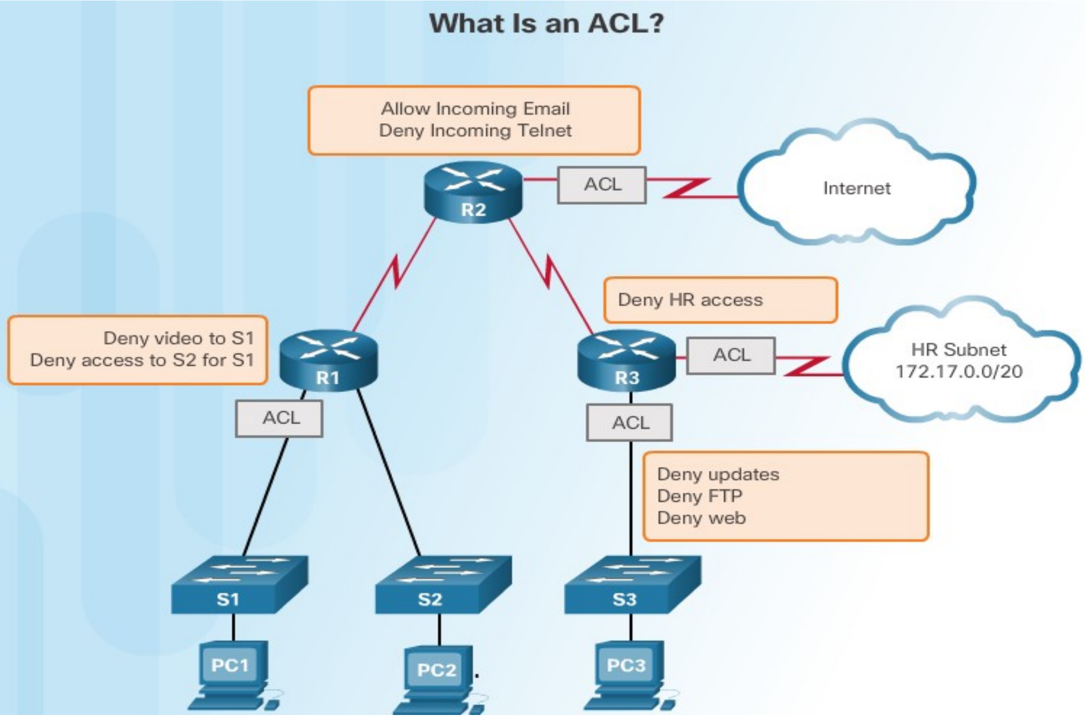




NAT Table with Overload

| Inside Global IP Address | Inside Local IP Address | Outside Local IP Address | Outside Global IP Address |
|---|---|---|---|
| 209.165.200.226:1555 | 192.168.10.10:1555 | 209.165.201.1:80 | 209.165.201.1:80 |
| 209.165.200.226:1331 | 192.168.10.11:1331 | 209.165.202.129:80 | 209.165.202.129:80 |

- R1(config)#access-list 1 permit 10.0.0.10 0.0.0.0
- R1(config)#access-list 1 permit 10.0.0.20 0.0.0.0
- R1(config)#ip nat pool ccna 50.0.0.1 50.0.0.1 netmask 255.0.0.0
- R1(config)#ip nat inside source list 1 pool ccna overload
- R1(config)#interface FastEthernet 0/0
- R1(config-if)#ip nat inside
- R1(config-if)#exit
- R1(config)#interface Serial 0/0/0
- R1(config-if)#ip nat outside

# What is an ACL?



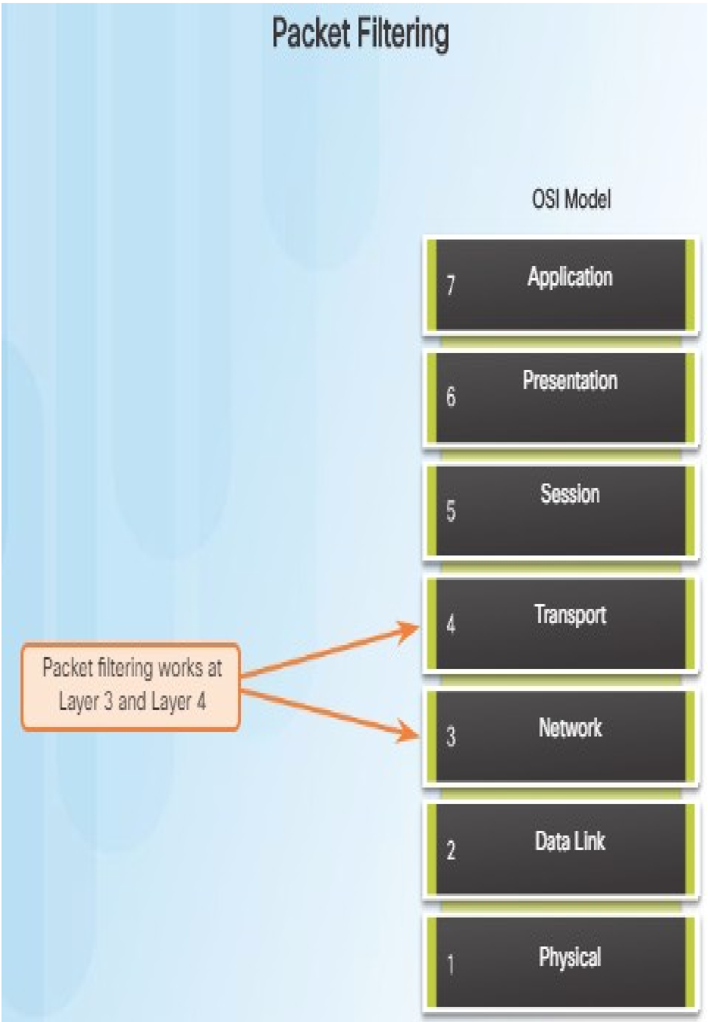

- An ACL is commands that control whether a router **forwards** or **drops packets** based on information found in the packet header.
- ACLs are not configured by default on a router.

# Cont…

- ACL's can perform the following tasks:
- **Limit network traffic** to increase network performance.

  **For example, video traffic** could be blocked if it's not permitted.
- **Provide traffic flow control**. ACLs can help **verify routing updates** are from a known source.
- **ACLs provide security** for network access and can **block a host or a network.**
- **Filter traffic based on traffic type** such as **Telnet** traffic.
- **Screen hosts to permit or deny access** to network services such as **FTP or HTTP**.

# Packet Filtering

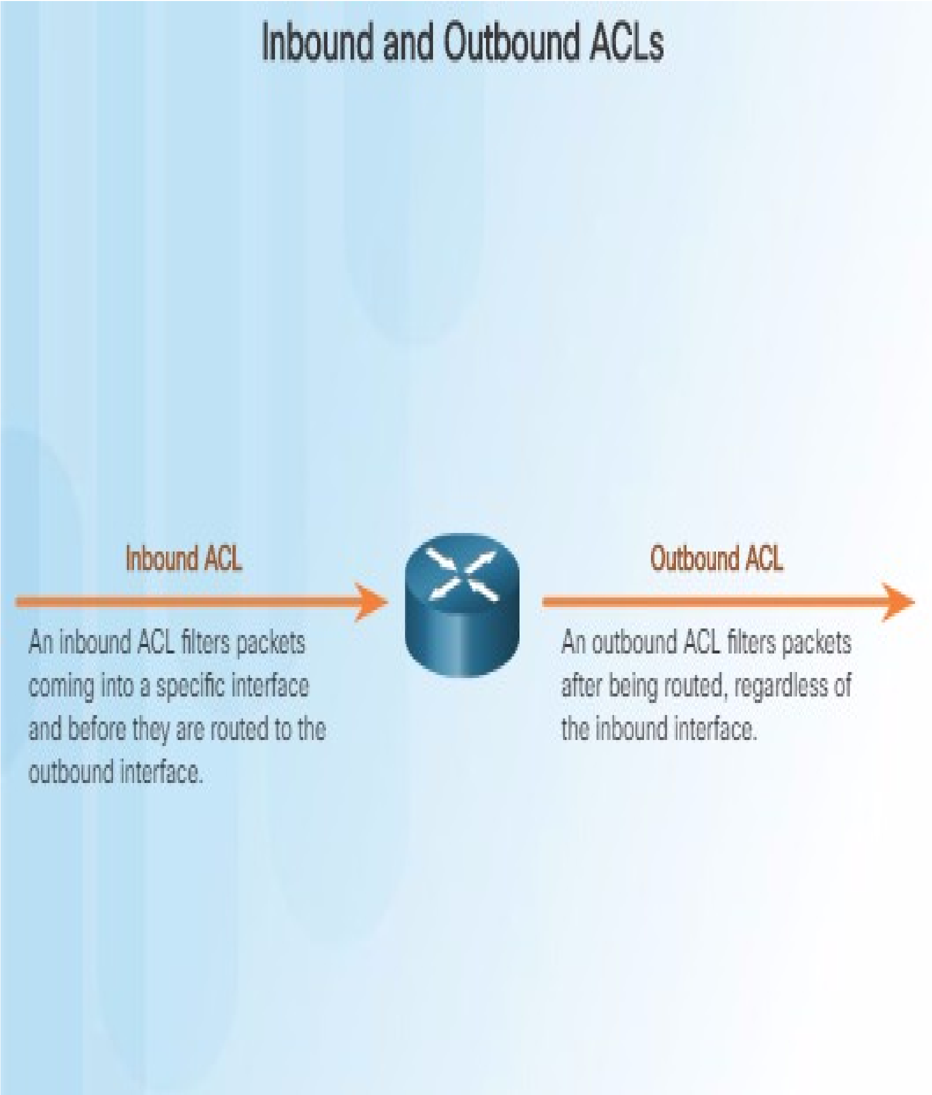



- When **network traffic passes** through an interface **configured with an ACL**, the **router compares** the information within the packet against each ACL, to determine if the packet matches one of the ACLs statements. This is referred to as packet filtering.

- Packet Filtering:

  Can analyze **incoming** and/or **outgoing packets**.

# ACL Operation

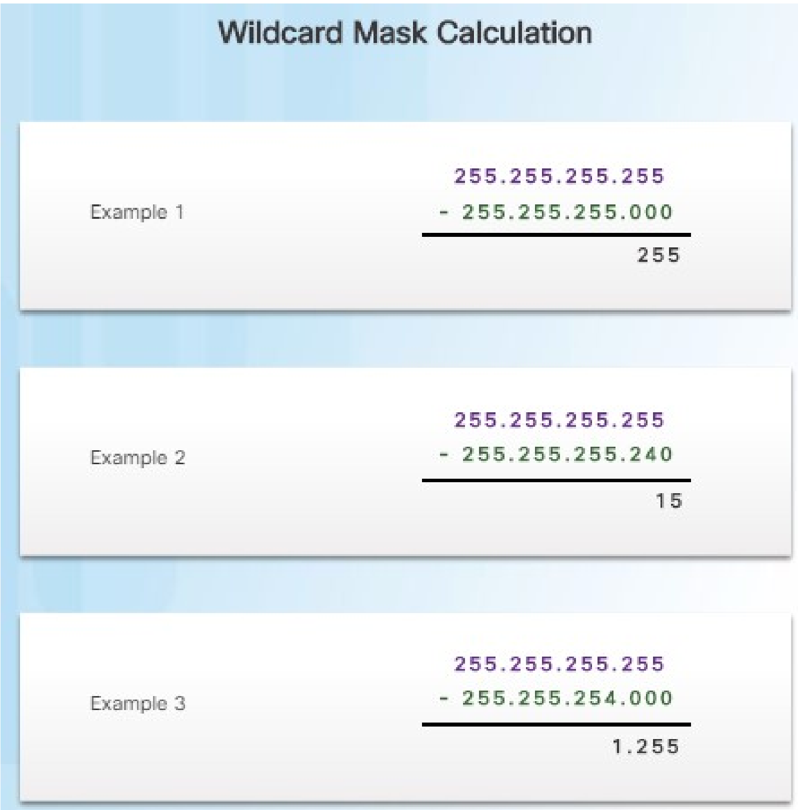



- ACLs can be configured to apply to **inbound traffic** and **outbound traffic:**

  **Inbound ACLs** – Incoming packets are processed before they are routed to the outbound interface.

  **Outbound ACLs** – Incoming packets are routed to the outbound interface, and then they are processed through the outbound ACL.

# Calculating the Wildcard Mask



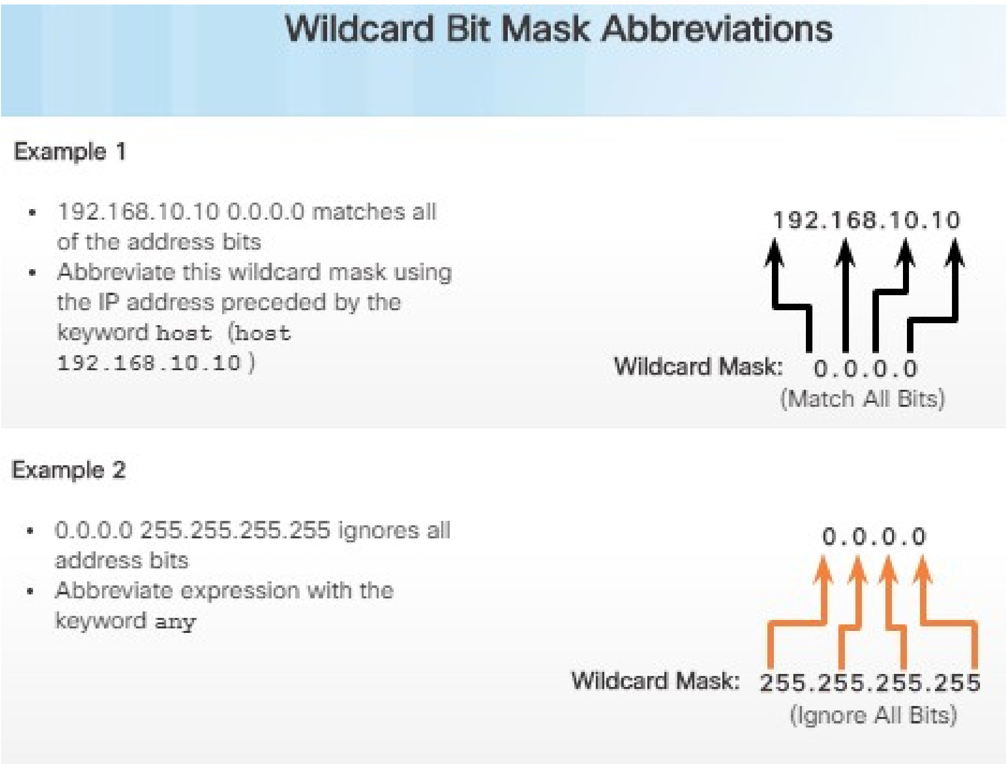

- Calculating wildcard mask examples:

Example 1: Assume you want to permit access to all users in the 192.168.3.0 network with the subnet mask of 255.255.255.0. Subtract the subnet from 255.255.255.255 and the result is: 0.0.0.255.

Example 2: Assume you want to permit network access for the 14 users in the subnet 192.168.3.32/28 with the subnet mask of 255.255.255.240. After subtracting the subnet maks from 255.255.255.255, the result is 0.0.0.15.

Example 3: Assume you want to match only networks 192.168.10.0 and 192.168.11.0 with the subnet mask of 255.255.254.0. After subtracting the subnet mask from 255.255.255.255, the result is 0.0.1.255.

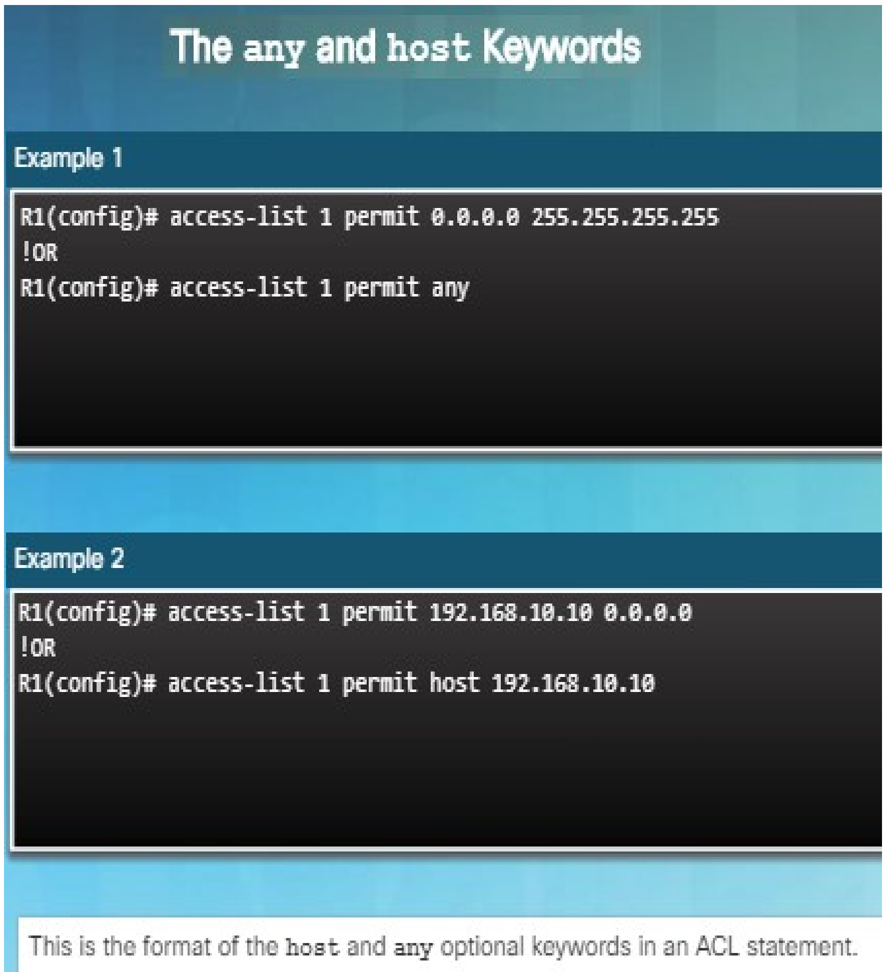
# Wildcard Mask Keywords




- To make wildcard masks easier to read, the keywords **host** and **any** can help identify the most common uses of wildcard masking.
  - **host** substitutes for the 0.0.0.0 mask
  - **any** substitutes for the 255.255.255.255 mask
- If you would like to match the 192.169.10.10 address, you could use **192.168.10.10 0.0.0.0** or, you can use: **host 192.168.10.10**
- In Example 2, instead of entering **0.0.0.0 255.255.255.255**, you can use the keyword **any** by itself.

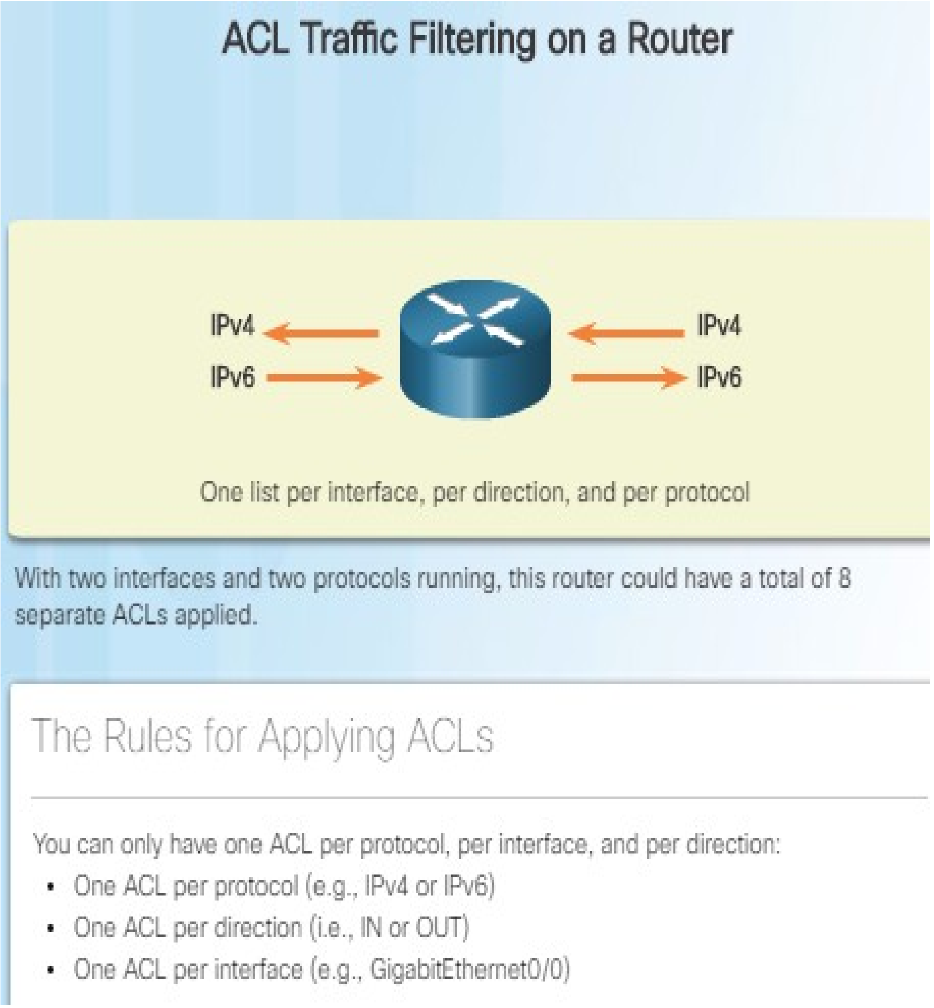
Wildcard Masks in ACLs

# Wildcard Mask Keyword Examples




- Example 1 in the figure demonstrates how to use the **any** keyword to substitute the IPv4 address 0.0.0.0 with a wildcard mask of 255.255.255.255.
- Example 2 demonstrates how to use the **host** keyword to substitute for the wildcard mask when identifying a single host.

# General Guidelines for Creating ACLs



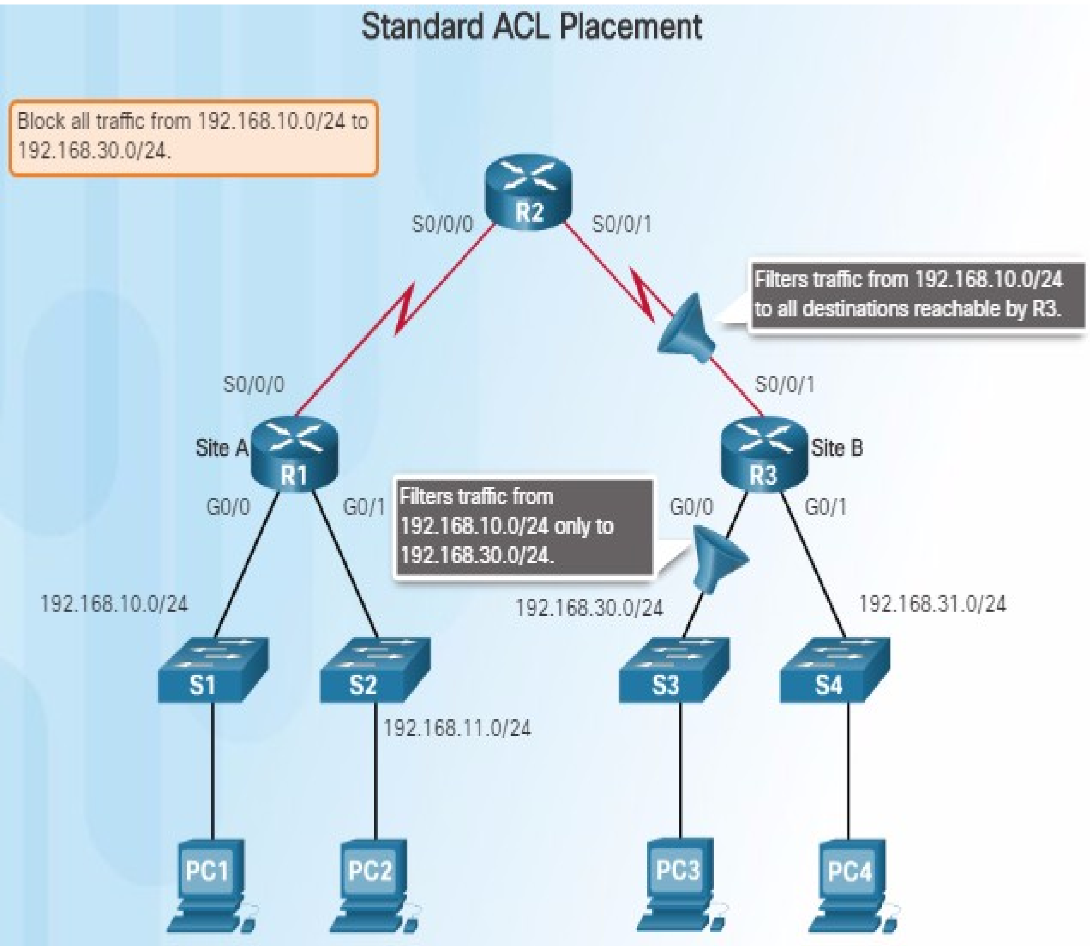

- Use ACLs in **firewall routers positioned between** your internal network and an external network such as the Internet.
- Use ACLs on a **router positioned between two parts of your network** to control traffic entering or exiting a specific part of your internal network.
- Configure ACLs on **border routers such as those situated at the edge** of your network. This will provide a basic buffer from the outside network that is less controlled.
- Configure ACLs for each **network protocol configured** on the border router interfaces.

# ACL…

- There are three types of ACLs:

1. **Standard ACLs**
2. **Extended ACLs**
3. **Named ACLs**

# Standard ACL Placement




- This example demonstrates the proper placement of the standard ACL that is configured to block traffic from the 192.168.10.0/24 network to the 192.168.30.0/24 network.
- There are two possible places to configure the access-list on R3.
- If the access-list is applied to the S0/0/1 interface, it will block traffic to the 192.168.30.0/24 network, **but also**, going to the 192.168.31.0/24 network.
- The **best place** to apply the access list is on **R3's G0/**0 interface. The access-list list should be applied to traffic exiting the G0/0 interface. Packets from 192.168.10.0/24 can still reach 192.168.31.0/24.

# Numbered Standard ACL

| Parameter | Description |
|---|---|
| *access-list-number* | Number of an ACL. This is a decimal number from 1 to 99, or 1300 to 1999 (for standard ACL). |
| **deny** | Denies access if the conditions are matched. |
| **permit** | Permits access if the conditions are matched. |
| **remark** | Add a remark about entries in an IP access list to make the list easier to understand and scan. |
| *source* | Number of the network or host from which the packet is being sent. There are two ways to specify the *source*:<br>• Use a 32-bit quantity in four-part, dotted-decimal format.<br>• Use the keyword **any** as an abbreviation for a *source* and *source-wildcard* of 0.0.0.0 255.255.255.255. |
| *source-wildcard* | (Optional) 32-bit wildcard mask to be applied to the source. Places ones in the bit positions you want to ignore. |
| **log** | (Optional) Causes an informational logging message about the packet that matches the entry to be sent to the console. (The level of messages logged to the console is controlled by the **logging console** command.)<br><br>The message includes the ACL number, whether the packet was permitted or denied, the source address, and the number of packets. The message is generated for the first packet that matches, and then at five-minute intervals, including the packet number of packets permitted or denied in the prior five-minute interval. |

- The **access-list** global configuration command defines a standard ACL with a number in the range of **1 through 99.**
- The full syntax of the standard ACL command is as follows:

  Router(config)# **access-list** *access-list-number* { **deny** | **permit** | **remark** } *source* [ *source-wildcard* ][ **log** ]

  To remove the ACL, the global configuration **no access-list** command is used. Use the **show access-list** command to verify the removal of the ACL.

# Applying Standard IPv4 ACLs to Interfaces

Step 1: Use the **access-list** global configuration command to create an entry in a standard IPv4 ACL.

```
R1(config)# access-list 1 permit 192.168.10.0 0.0.0.255
```

The example statement matches any address that starts with 192.168.10.x. Use the **remark** option to add a description to your ACL.

Step 2: Use the **interface** configuration command to select an inteface to which to apply the ACL.

```
R1(config)# interface serial 0/0/0
```

Step 3: Use the **ip access-group** interface configuration command to activate the existing ACL on an interface.

```
R1(config-if)# ip access-group 1 out
```

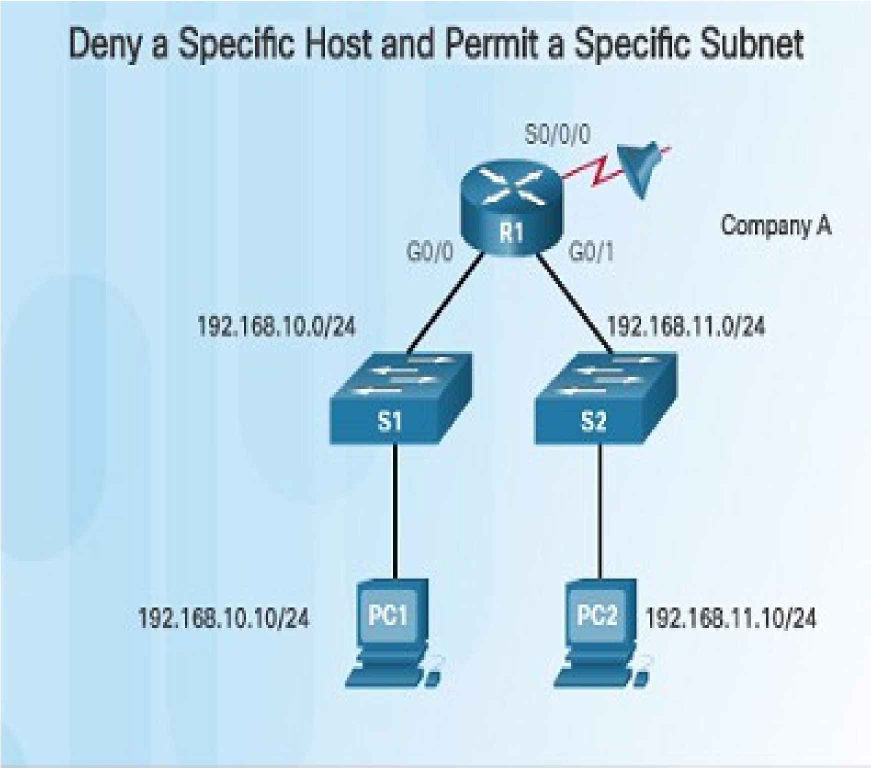
This example activates the standard IPv4 ACL 1 on the interface as an outbound filter.

- After a standard IPv4 ACL is configured, it is linked to an interface using the **ip access-group** command in interface configuration mode:

  Router(config-if)# **ip access-group** { *access-list-number* | *access-list-name* } { **in** | **out** }

- To remove an ACL from an interface, first enter the **no ip access-group** command on the interface, and then enter the global **no access-list** command to remove the entire ACL.

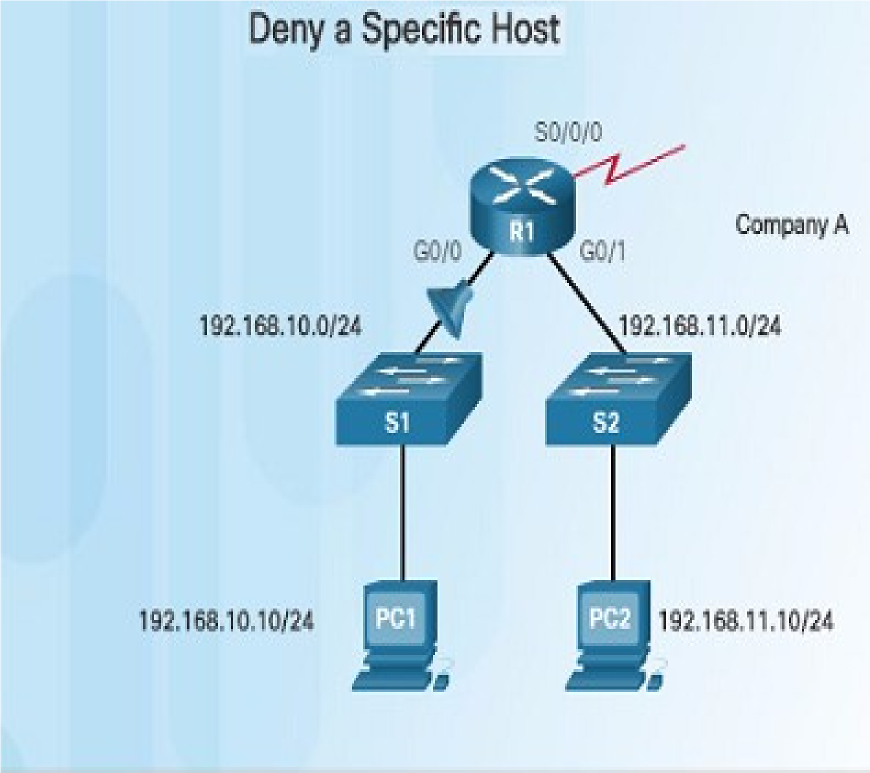
# Numbered Standard IPv4 ACL Examples




```
R1(config)# no access-list 1
R1(config)# access-list 1 deny host 192.168.10.10
R1(config)# access-list 1 permit 192.168.10.0 0.0.0.255
R1(config)# interface s0/0/0
R1(config-if)# ip access-group 1 out
```

- The figure to the left shows an example of an ACL that permits traffic from a specific subnet but denies traffic from a specific host on that subnet.
  - The **no access-list 1** command deletes the previous version of ACL 1.
  - The next ACL statement denies the host 192.168.10.10.
  - What is another way to write this command without using **host**?
  - All other hosts on the 192.168.10.0/24 network are then permitted.
  - There is an implicit deny statement that matches every other network.
  - Next, the ACL is reapplied to the interface in an outbound direction.

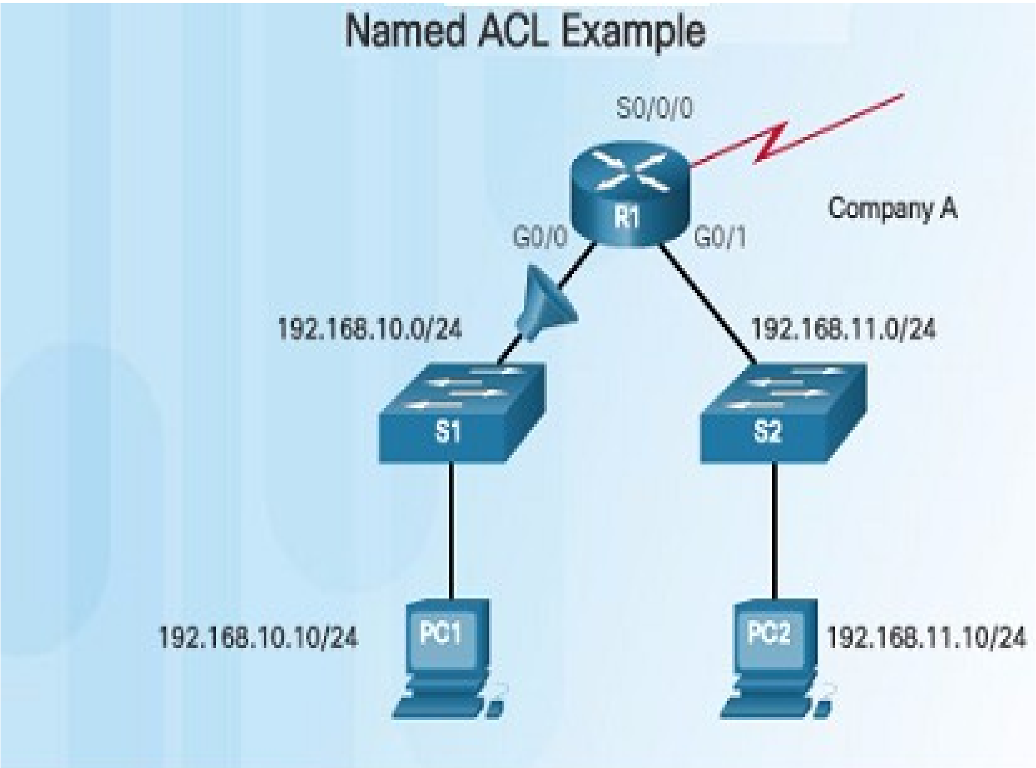
# Numbered Standard ACL




```
R1(config)# no access-list 1
R1(config)# access-list 1 deny host 192.168.10.10
R1(config)# access-list 1 permit any
R1(config)# interface g0/0
R1(config-if)# ip access-group 1 in
```

- This next example demonstrates an ACL that denies a specific host but will permit all other traffic.
  - The first ACL statement deletes the previous version of ACL 1.
  - The next command, with the deny keyword, will deny traffic from the PC1 host that is located at 192.168.10.10.
  - The **access-list 1 permit any** statement will permit all other hosts.
  - This ACL is applied to interface G0/0 in the inbound direction since it only affects the 192.168.10.0/24 LAN.

# Named Standard IPv4 ACL Syntax




```
R1(config)# ip access-list standard NO_ACCESS
R1(config-std-nacl)# deny host 192.168.11.10
R1(config-std-nacl)# permit any
R1(config-std-nacl)# exit
R1(config)# interface g0/0
R1(config-if)# ip access-group NO_ACCESS out
```

- Identifying an ACL with a name rather than with a number makes it easier to understand its function.
- The example to the left shows how to configured a named standard access list. Notice how the commands are slightly different:
  - Use the **ip access-list** command to create a named ACL. Names are alphanumeric, case sensitive, and must be unique.
  - Use permit or deny statements as needed. You can also use the **remark** command to add comments.
  - Apply the ACL to an interface using the **ip access-group** *name* command.

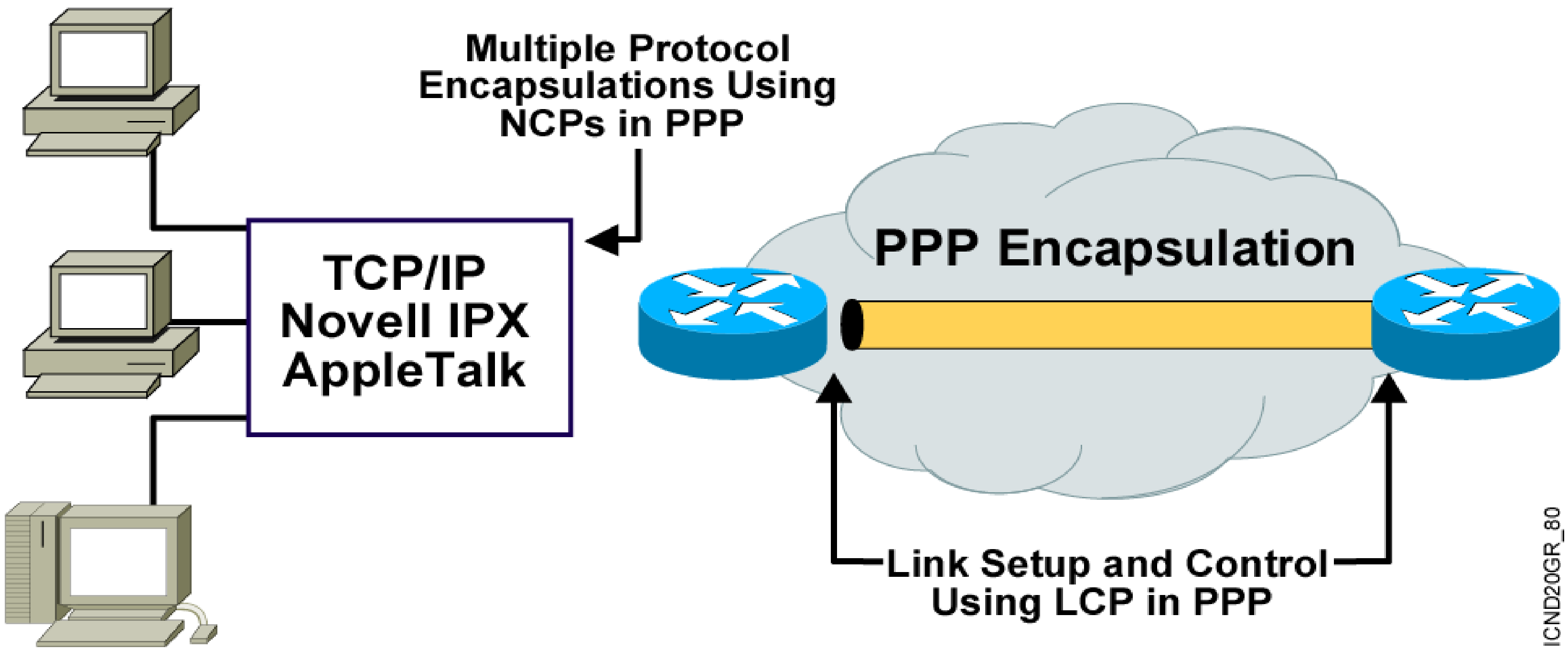
# Extended ACLs

- Extended ACLs filter not only on the source IP address but also on the **destination IP address**, **protocol**, and **port numbers**.

- The **range** of numbers for Extended ACLs is from **100 to 199** and from **2000 to 2699**.

Router(config)#access-list 100 deny tcp 172.16.2.0 0.0.0.255 any eq telnet

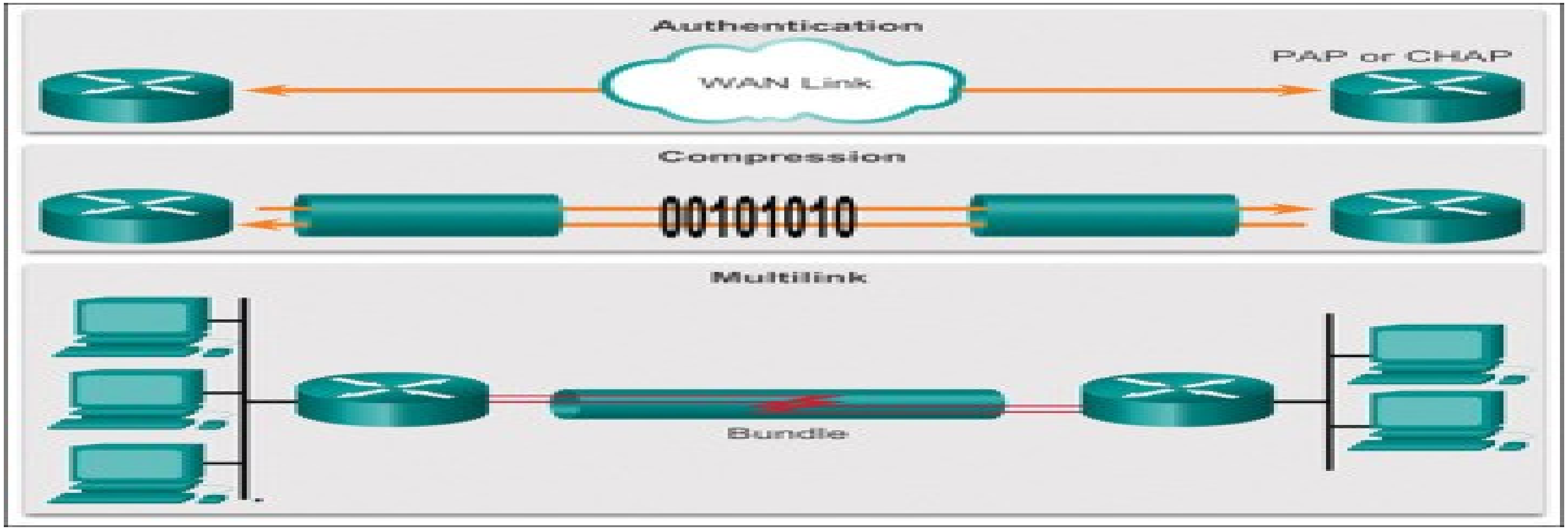
# Point to Point Protocol




- PPP can **carry packets from several protocol** suites using NCP(Network Control Protocol).
- PPP **controls** the setup of several link options using **LCP(**Link control protocol).

# Cont….

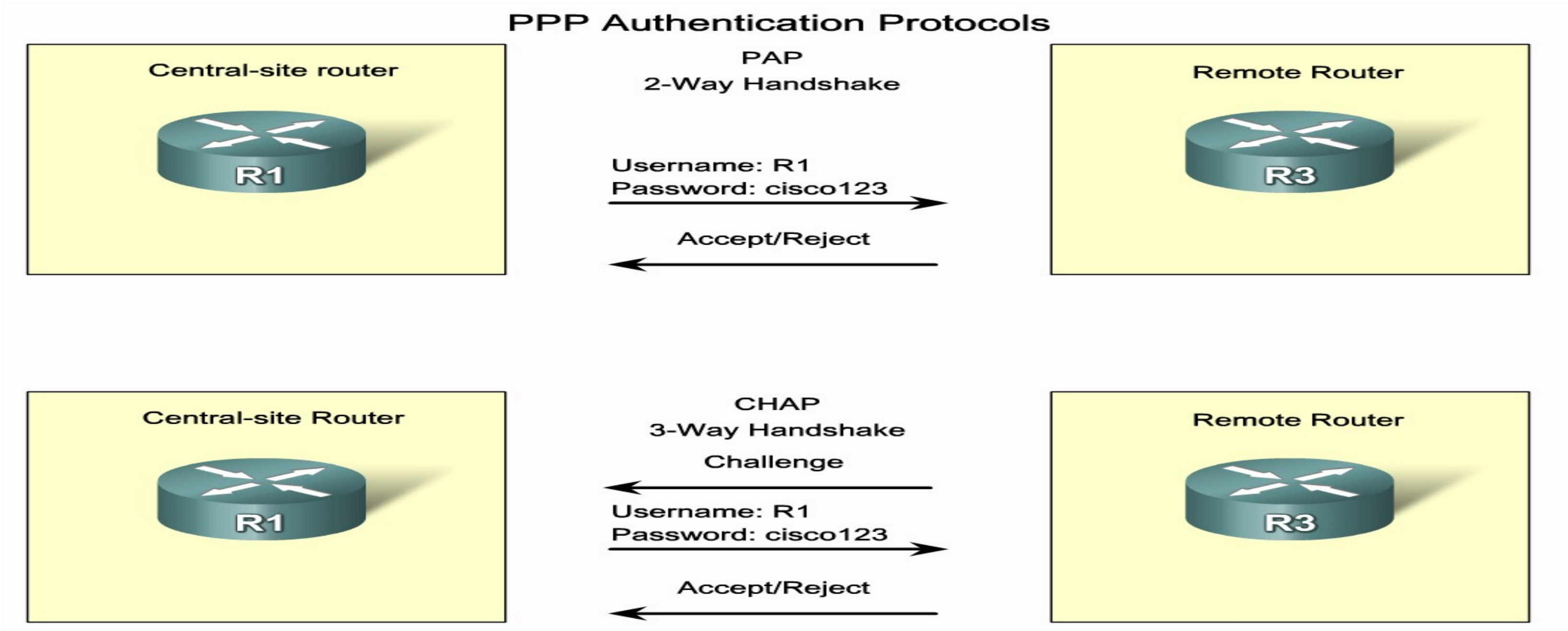
- **HDLC** is the Cisco default data-link layer protocol for **encapsulating data on synchronous serial data links.**
- **PPP** encapsulates network layer protocol information over **point-to-point links**.
- Configurable aspects of PPP include methods of **authentication**, **compression**, and, as well as **whether or not multilink is supported**.



# Cont…

- PPP session establishment progresses through three phases: **link establishment**, **authentication,** and **network layer protocol.**

- When configuring **PPP authentication**, you can select PAP or **CHAP**. In general, CHAP is the preferred protocol.

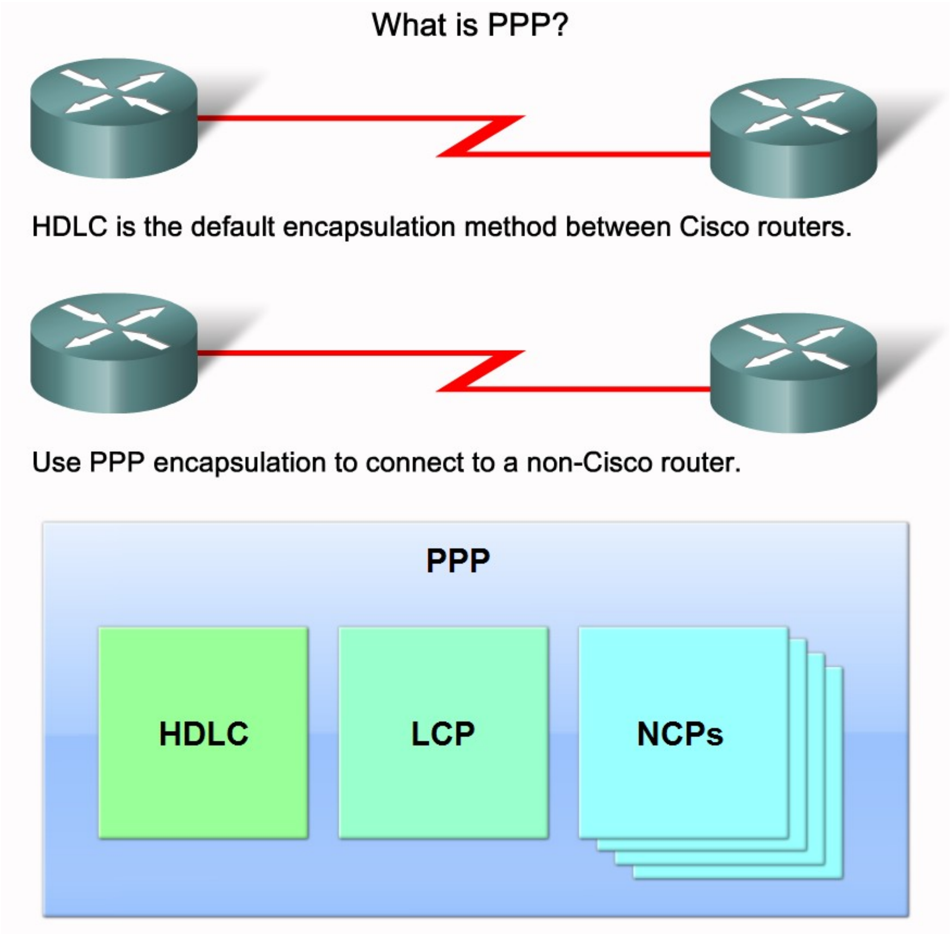


# Cont…

- You enable PPP with the **encapsulation ppp** command
- PPP authentication with the **ppp authentication** command.
- Use the **show interface** command to verify proper configuration of PPP encapsulation.
- The **debug ppp authentication** command displays the authentication exchange sequence.

# Point-to-Point protocol

- Describe PPP in terms of its use in WAN links

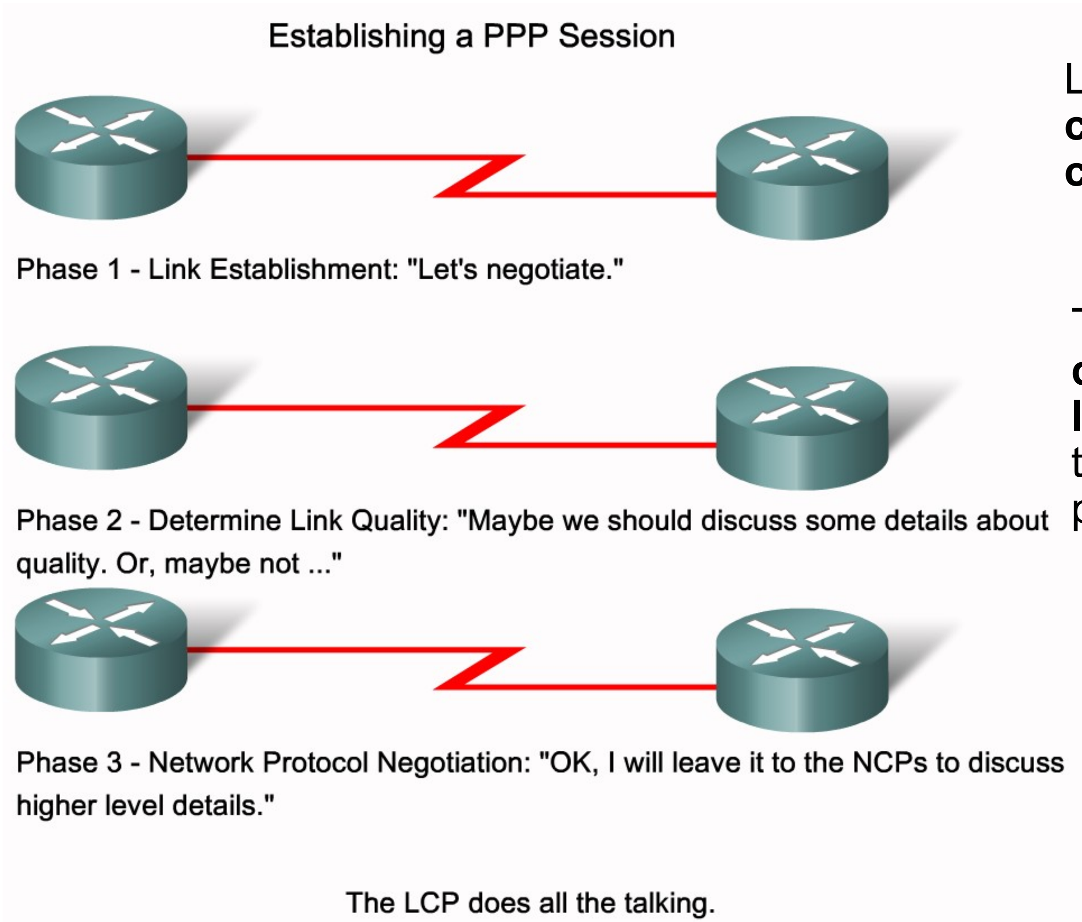



- The link quality management feature **monitors the quality of the link**. If too many errors are detected, PPP takes the link down.

- PPP supports PAP and CHAP authentication. This feature is explained and practiced in a later section.

# Point-to-Point protocol

- Define the three phases of PPP session establishment




LCP must first **open the connection** and **negotiate configuration options**.

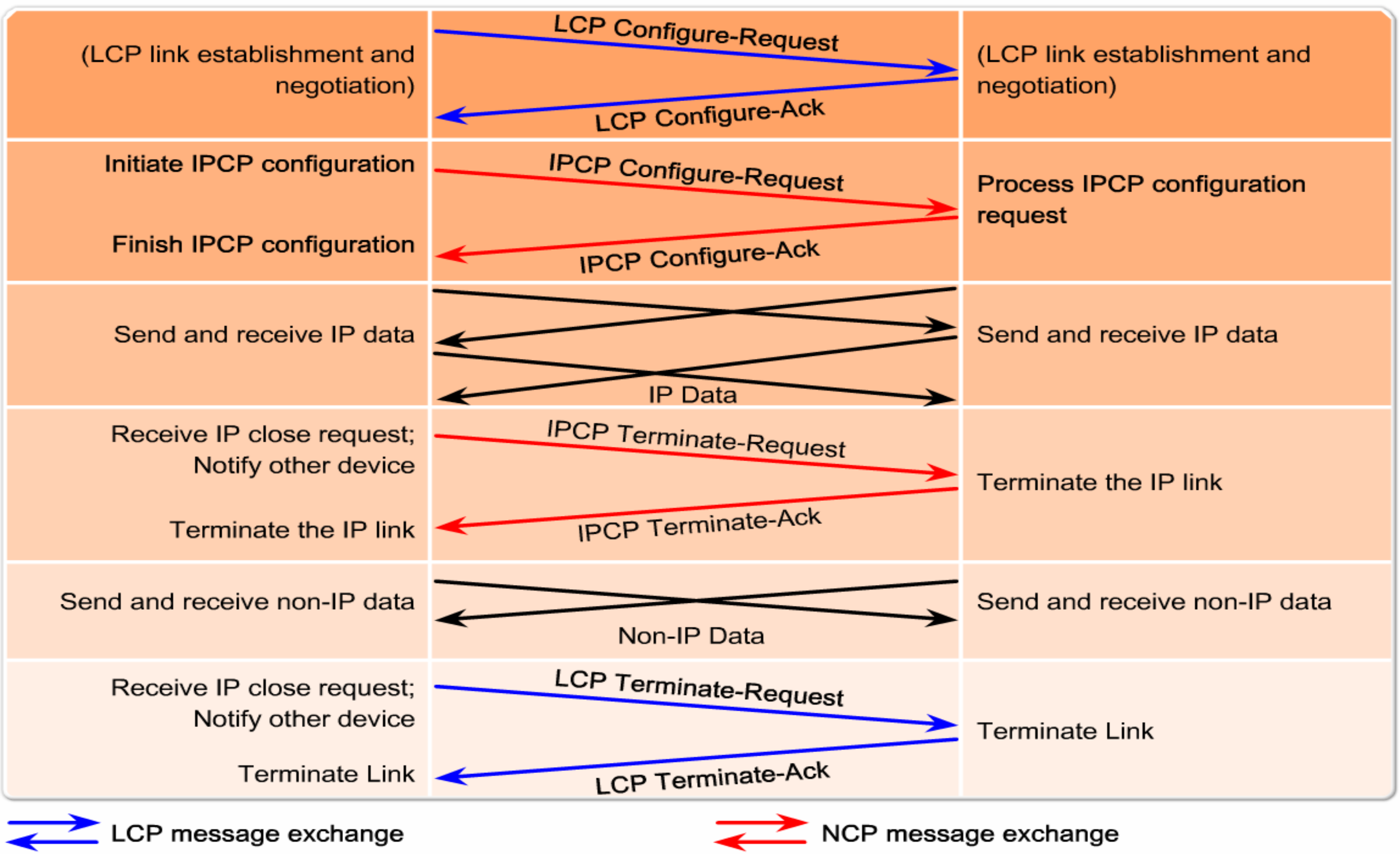
The **LCP tests the link to determine whether the link quality is sufficient** to bring up network layer protocols.

**NCP** can separately configure the network layer protocols, and bring them up and take them down at any time.

# Point-to-Point protocol

- Describe the characteristics of NCP

NCP Process



# Cont…

- **Enables HDLC encapsulation**
- Uses the default encapsulation on synchronous serial interfaces

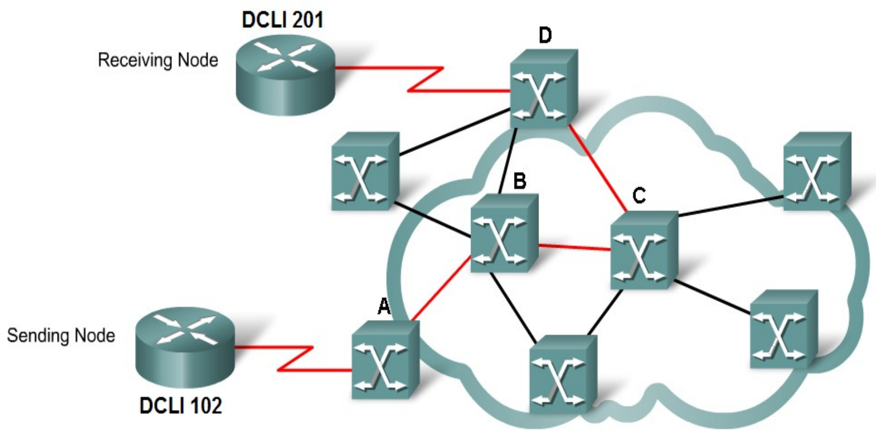
```
Router(config-if)#encapsulation hdlc
```

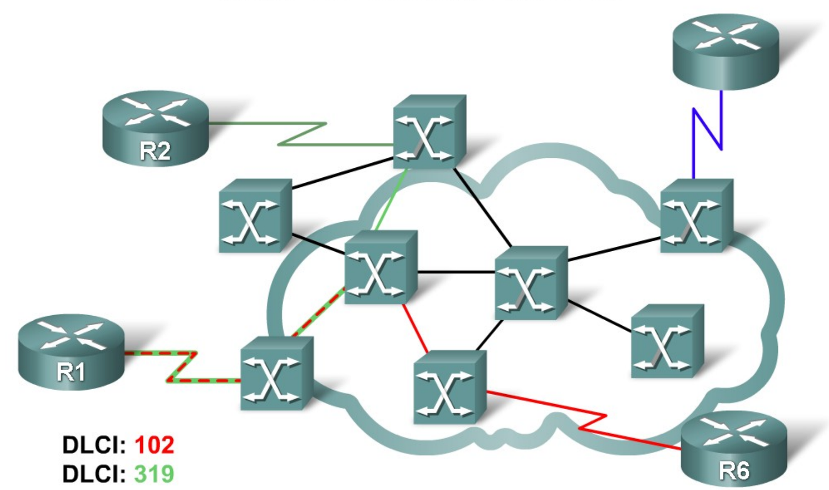
- **Enables PAP and/or CHAP authentication**

**Router(config-if)#ppp authentication {chap | chap pap | pap chap | pap}**

# Frame Relay

- Frame Relay allows for **a single serial interface** on the router to **connect multiple remote sites** with the help of **virtual circuits**.
- Although Frame Relay is popular due to its **relatively inexpensive** nature
- Describe how Frame Relay uses virtual circuits to carry packets from one DTE to another




Multiple VCs on the same access link are distinguishable by the DLCI.

# Cont…

- Frame relay is the most widely used WAN technology because it:
  - **Provides greater bandwidth** than leased line
  - **Reduces cost** because it uses less equipment
  - Easy to implement

- Frame relay is configured on virtual circuits
- **A permanent virtual circuit (PVC)** is a connection that is permanently **established between two or more nodes** in **frame relay**
- These **virtual circuits** may be identified by a **DLCI**

# Configure a Basic Frame Relay PVC

- Configure a basic Frame Relay PVC on a router serial interface